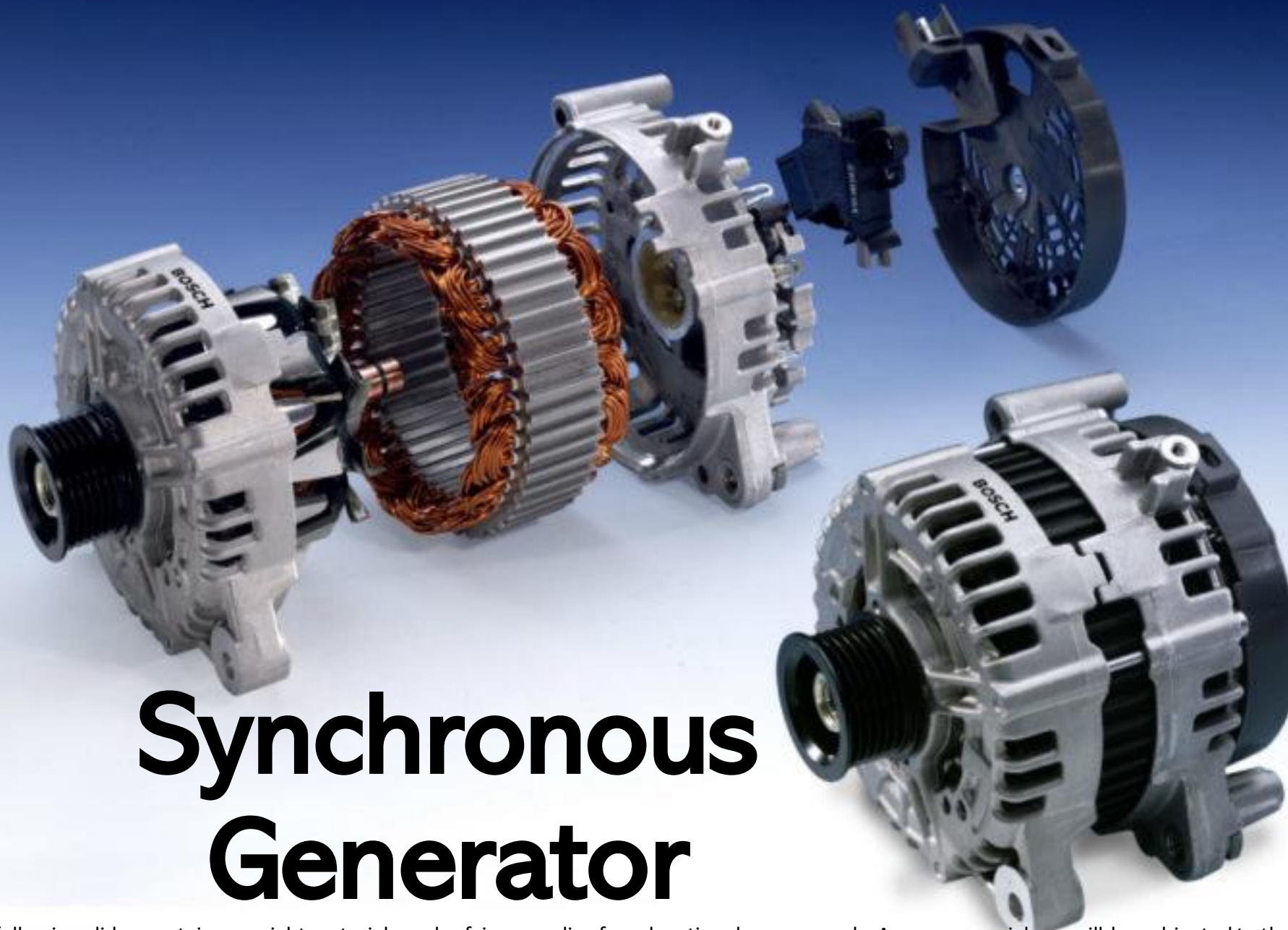




Synchronous Generator

History of Synchronous generator

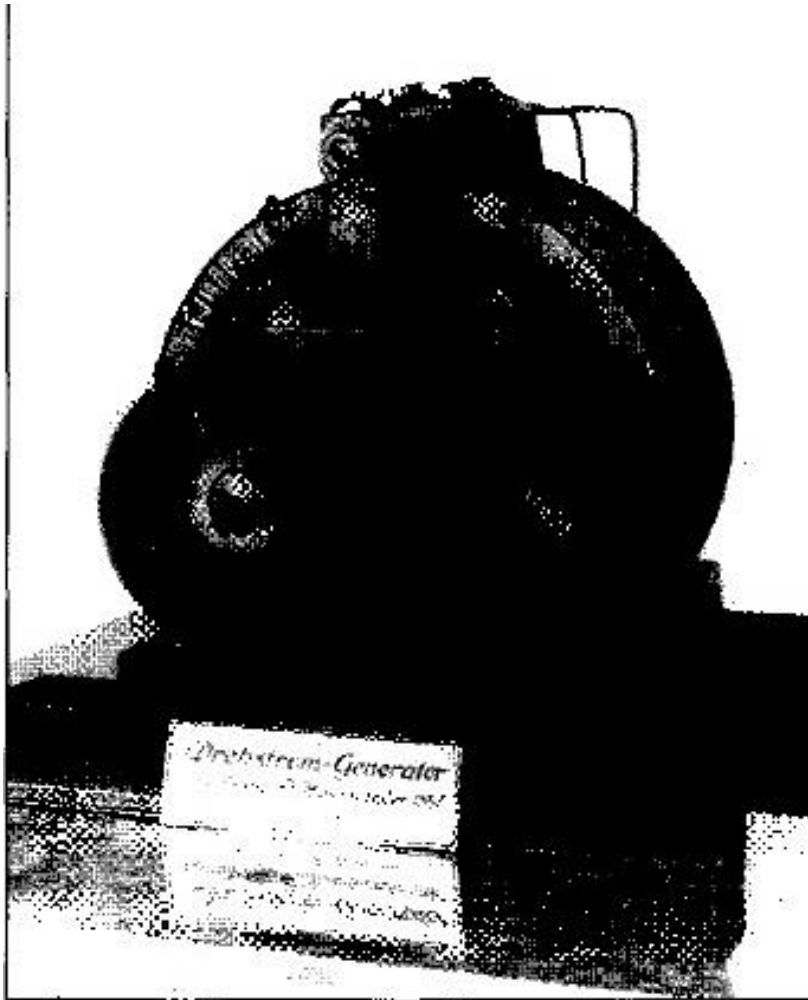


Fig. 1 The first three-phase synchronous machine, built by Friedrich August Haselwander in 1887 (Photo: Deutsches Museum, Munich)

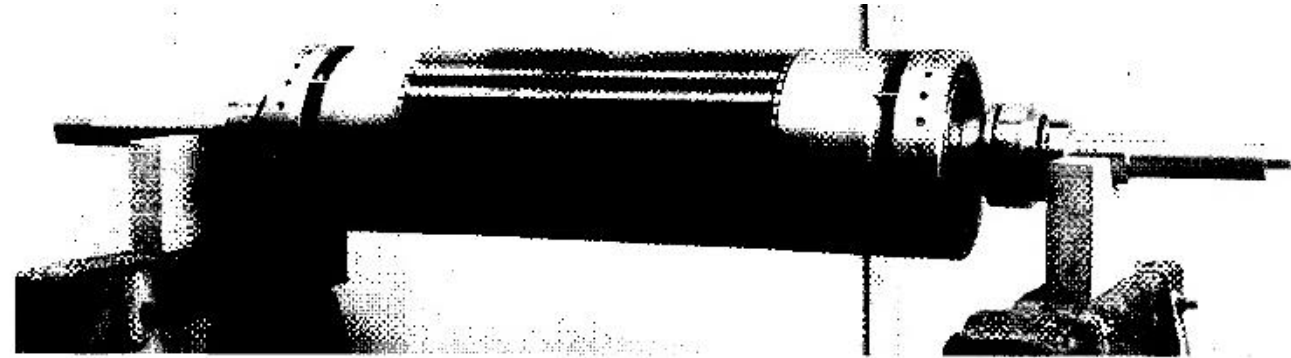
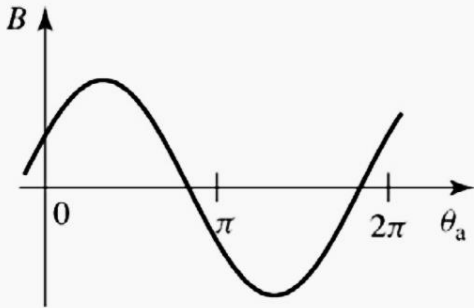
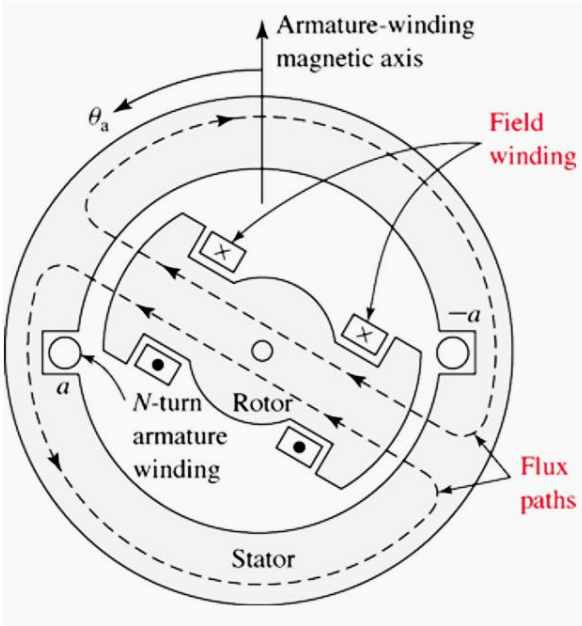


Fig. 8 The first two-pole turborotor in the world, built by Charles E. Brown in 1901

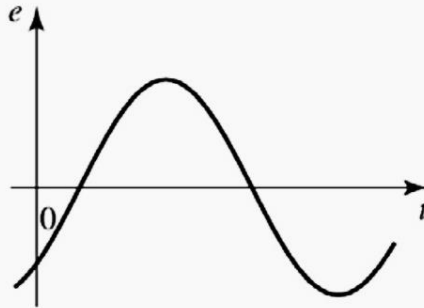


Fig. 7 First turborotor in the world, 100 kVA, 6-pole, built by Charles E. Brown in the year 1898

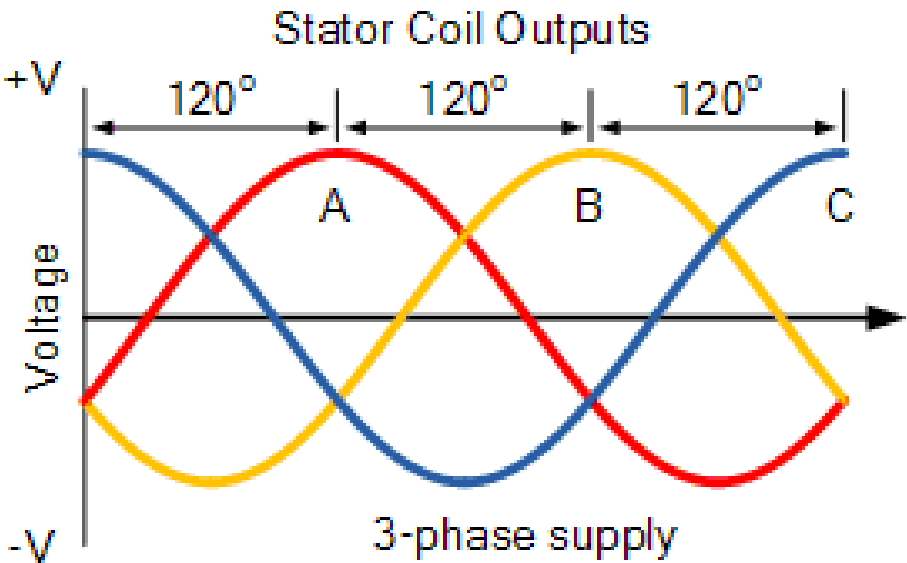
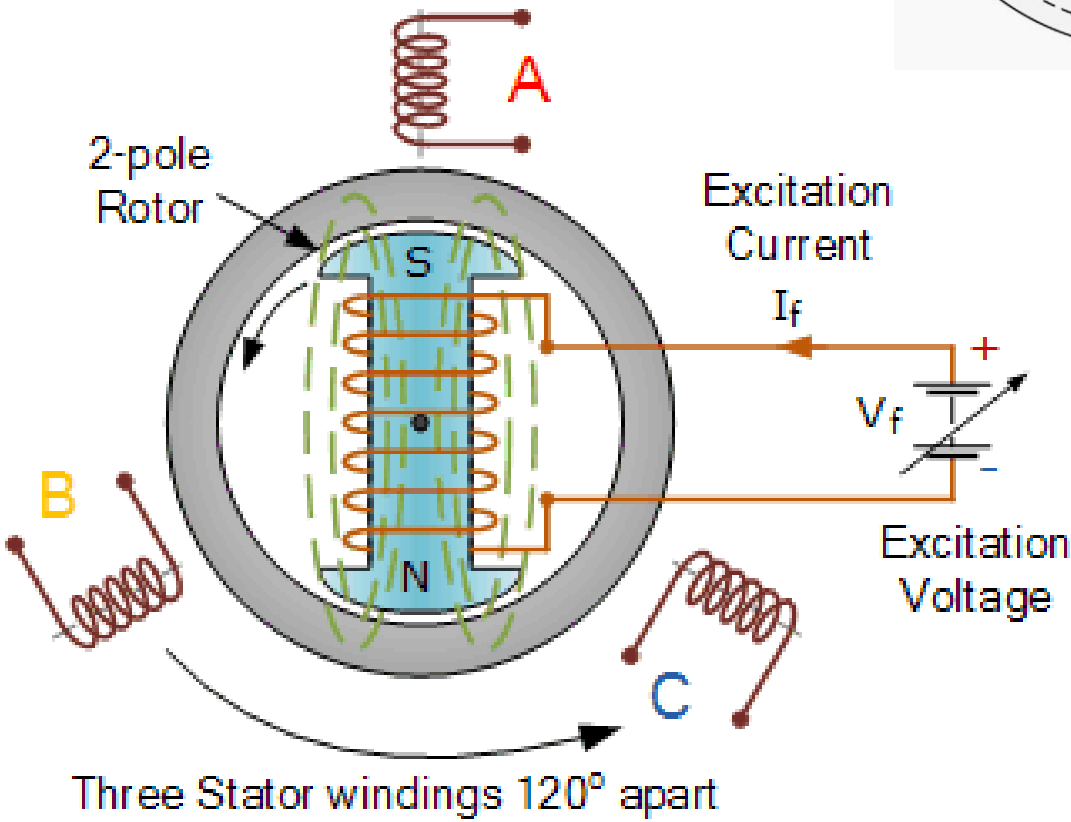
Alternator (5.1)



(a)



(b)



Video links for SG working principle

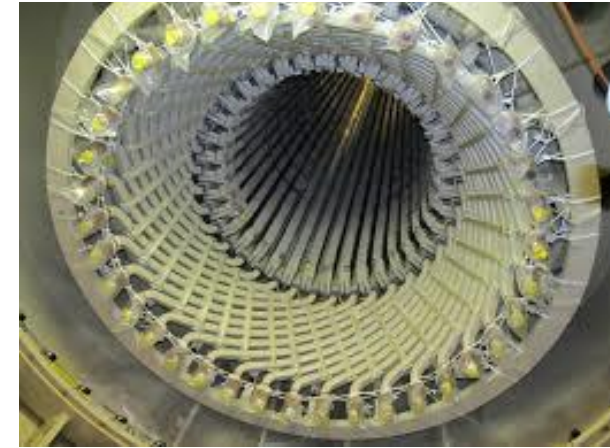
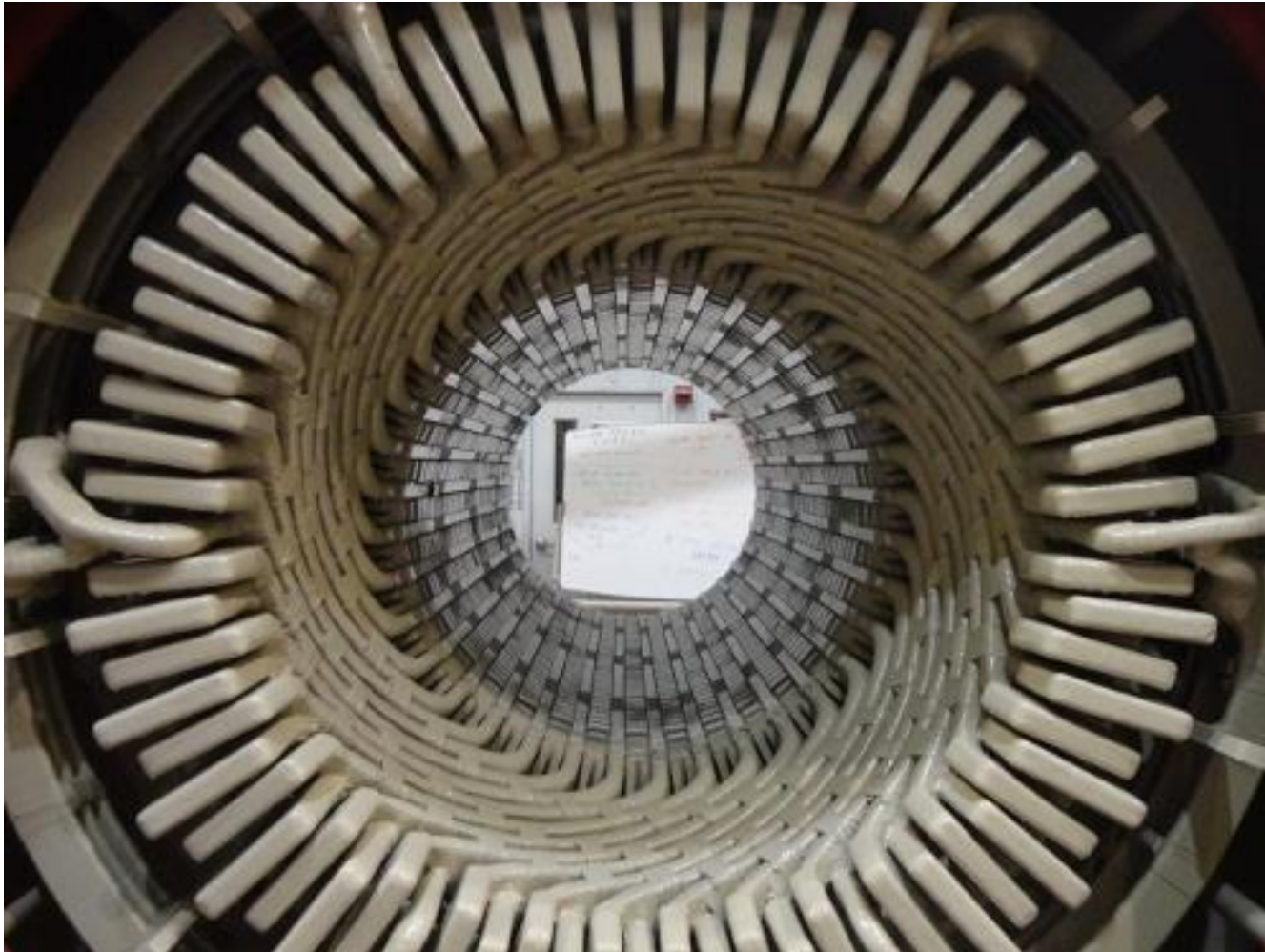
1. <https://www.youtube.com/watch?v=tiKH48EMgKE>
2. <https://www.youtube.com/watch?v=Ak6Zf-wJpsM>

Youtube channel (FYI) : ABB Motors and generators

<https://www.youtube.com/user/ABBmotorsgenerators>

Construction (5.1)

1. Stator:

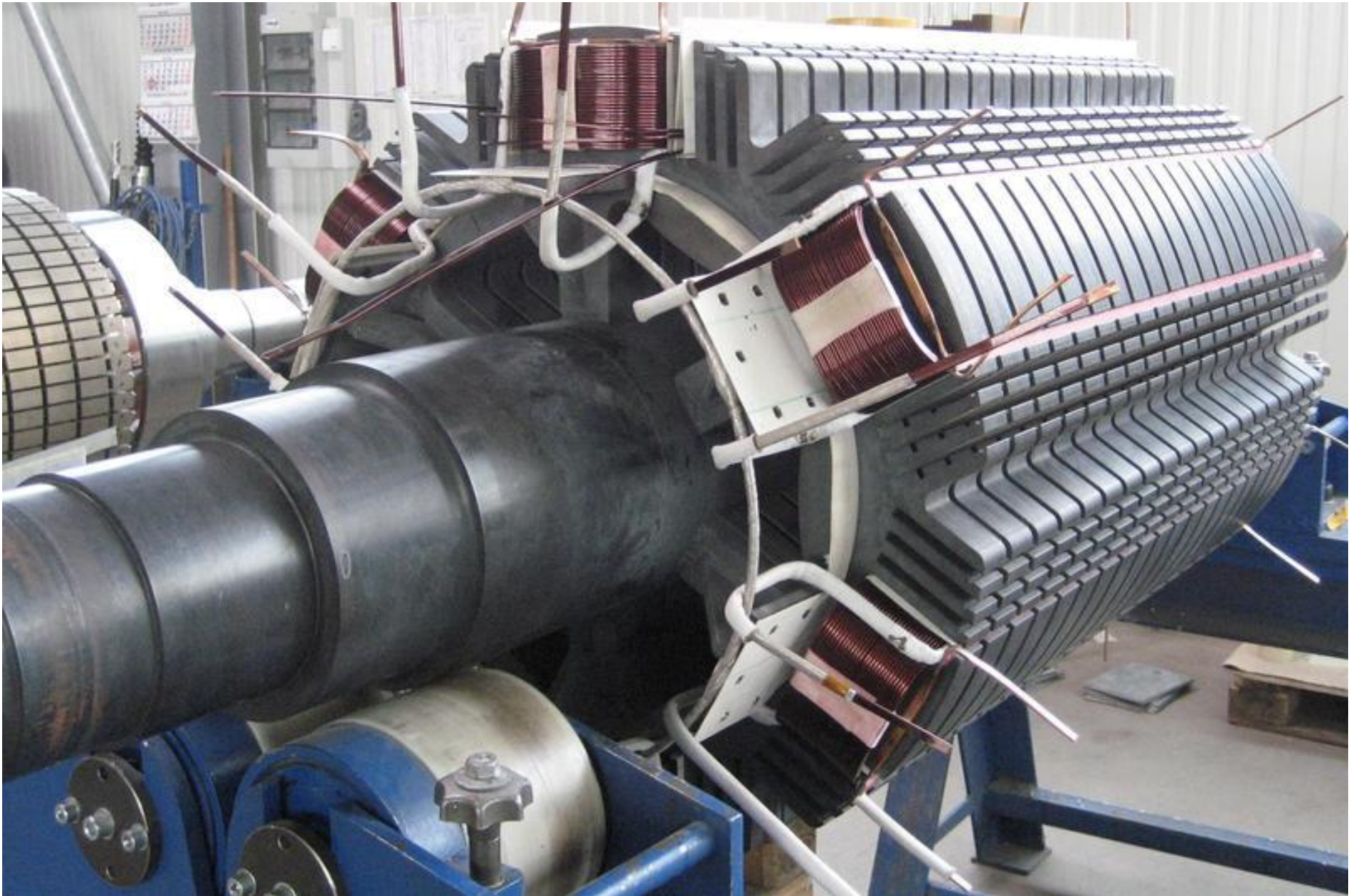




Stator rewinding:

https://www.youtube.com/watch?v=yEBuR__q9M0&ab_channel=MD%26ATurbines

2. Rotor



A. Salient (projecting) Pole

2. Rotor



B. Non-salient Pole

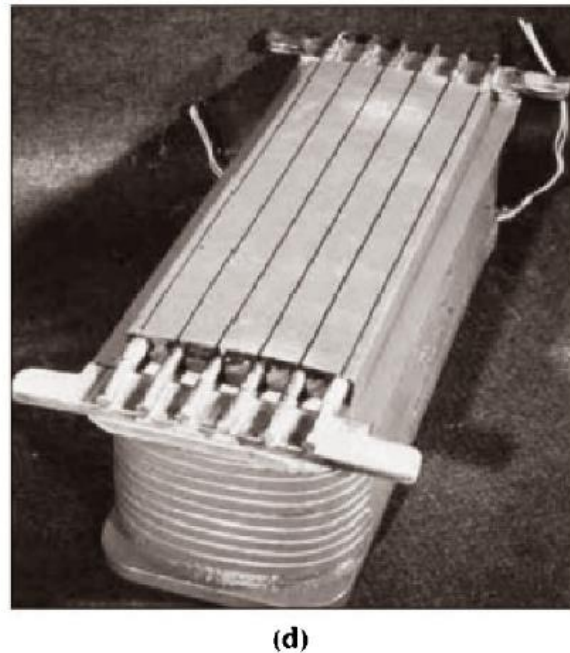
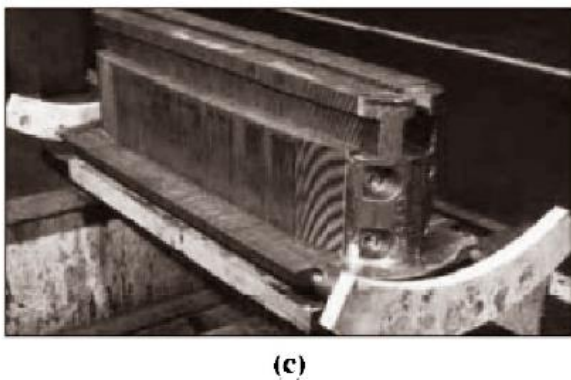
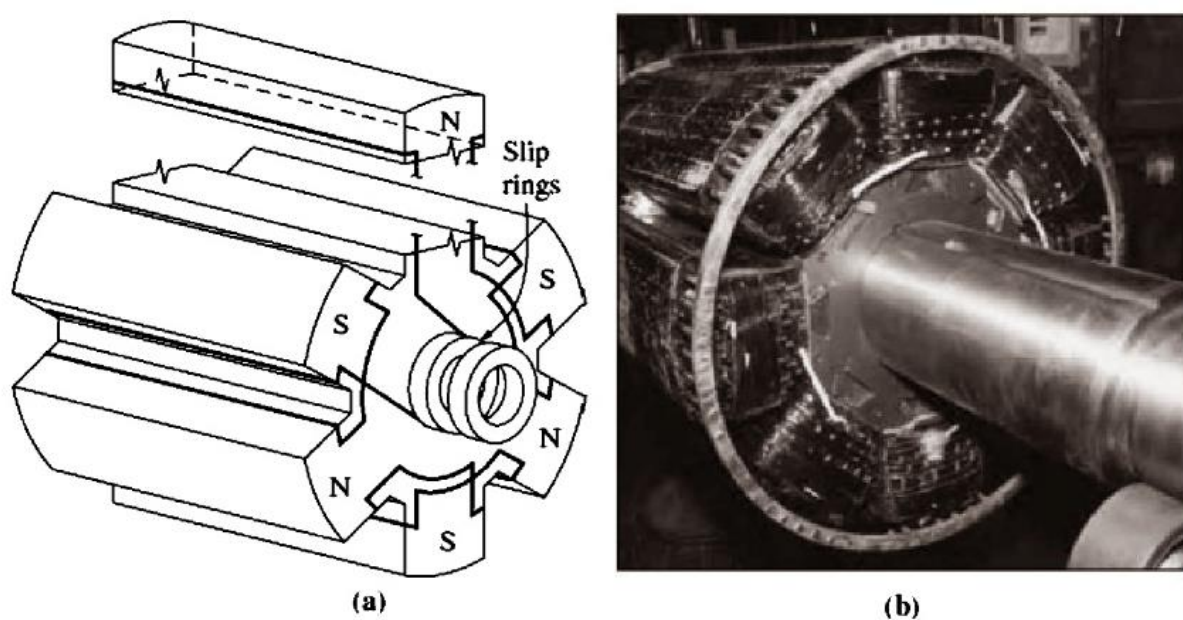


FIGURE 5-2
 (a) A salient six-pole rotor for a synchronous machine. (b) Photograph of a salient eight-pole synchronous machine rotor showing the windings on the individual rotor poles. (Courtesy of General Electric Company.) (c) Photograph of a single salient pole from a rotor with the field windings not yet in place. (Courtesy of General Electric Company.) (d) A single salient pole shown after the field windings are installed but before it is mounted on the rotor. (Courtesy of Westinghouse Electric Company.)

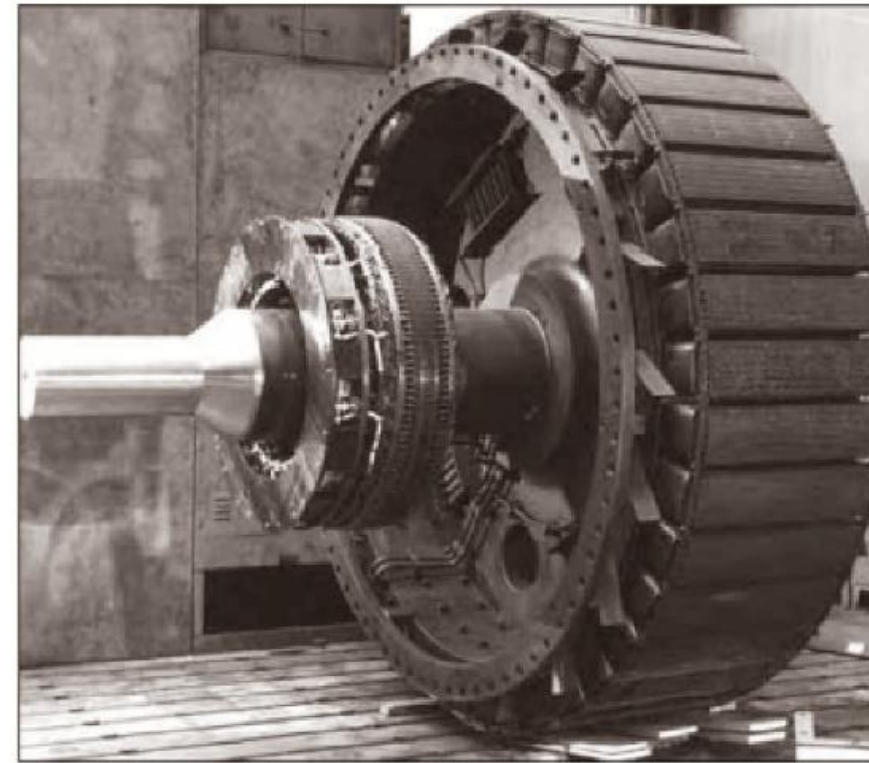
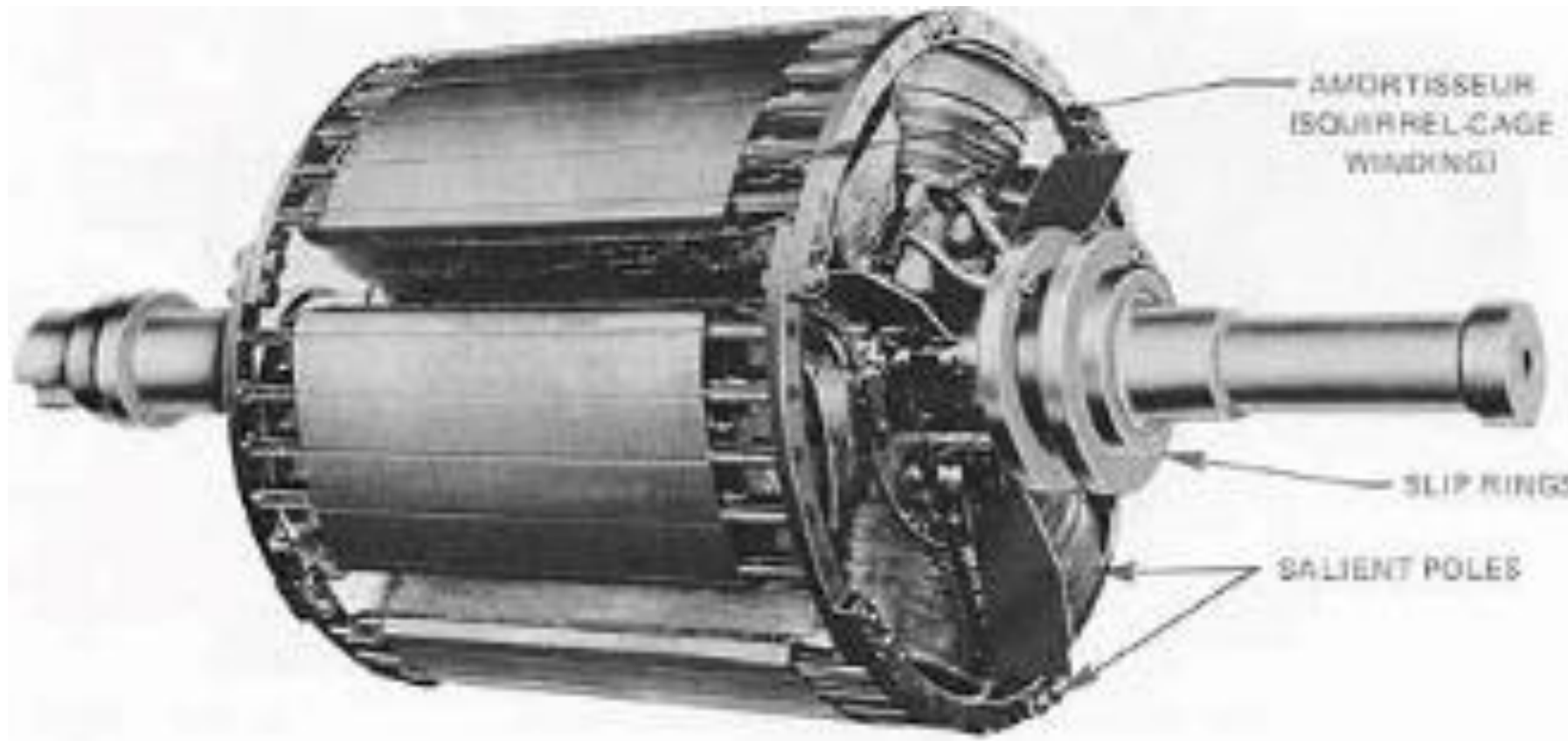


FIGURE 5-4
 Photograph of a synchronous machine rotor with a brushless exciter mounted on the same shaft. Notice the rectifying electronics visible next to the armature of the exciter. (Courtesy of Westinghouse Electric Company.)

How to supply dc power to rotor?

1. Supply the dc power from an external dc source to the rotor by means of slip rings and busbar
2. Supply the dc power from a special dc power source mounted on the shaft (brushless exciter)





Slip rings



Slip ring and brush

Brushless Exciter (5.1)

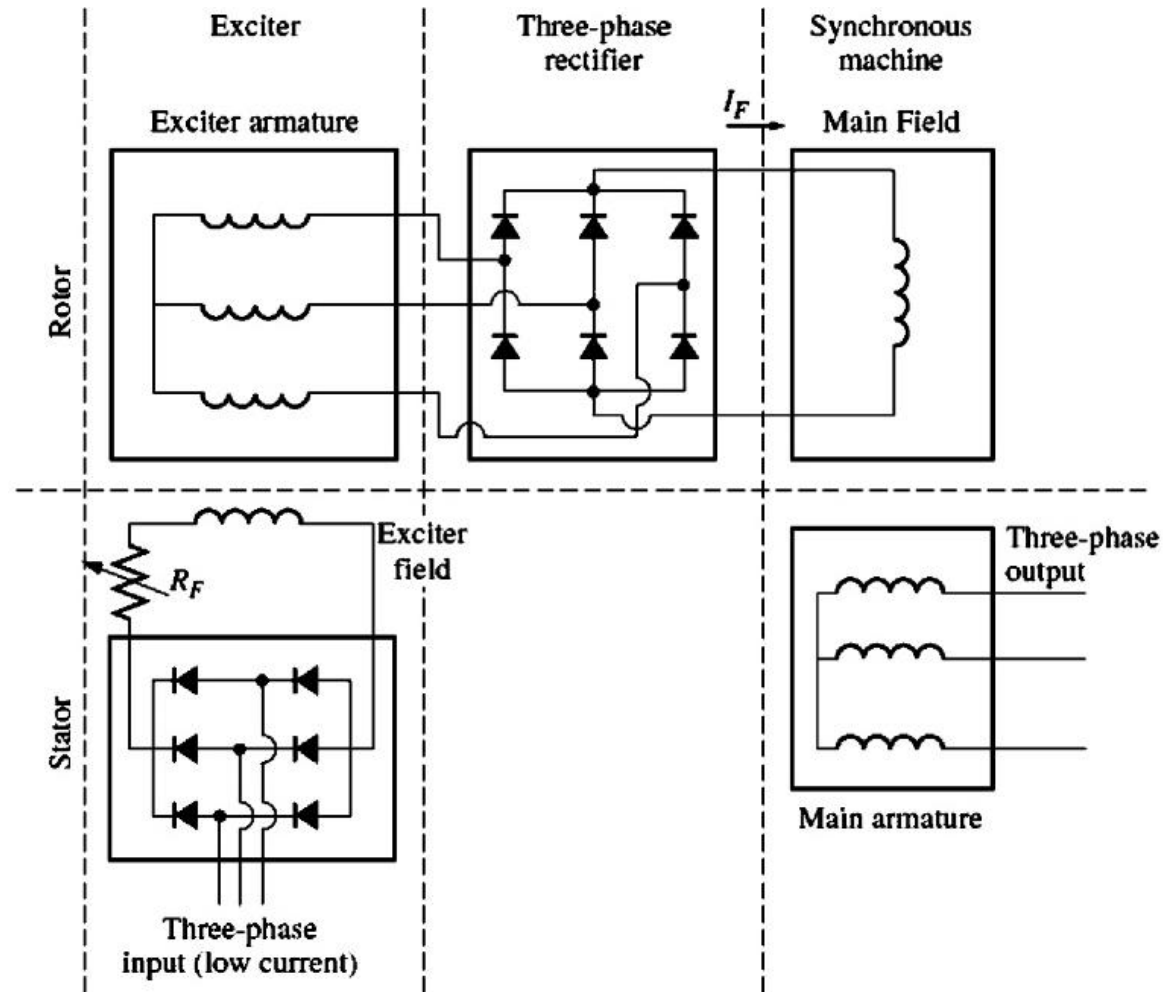


FIGURE 5-3

A brushless exciter circuit. A small three-phase current is rectified and used to supply the field circuit of the exciter, which is located on the stator. The output of the armature circuit of the exciter (on the rotor) is then rectified and used to supply the field current of the main machine.

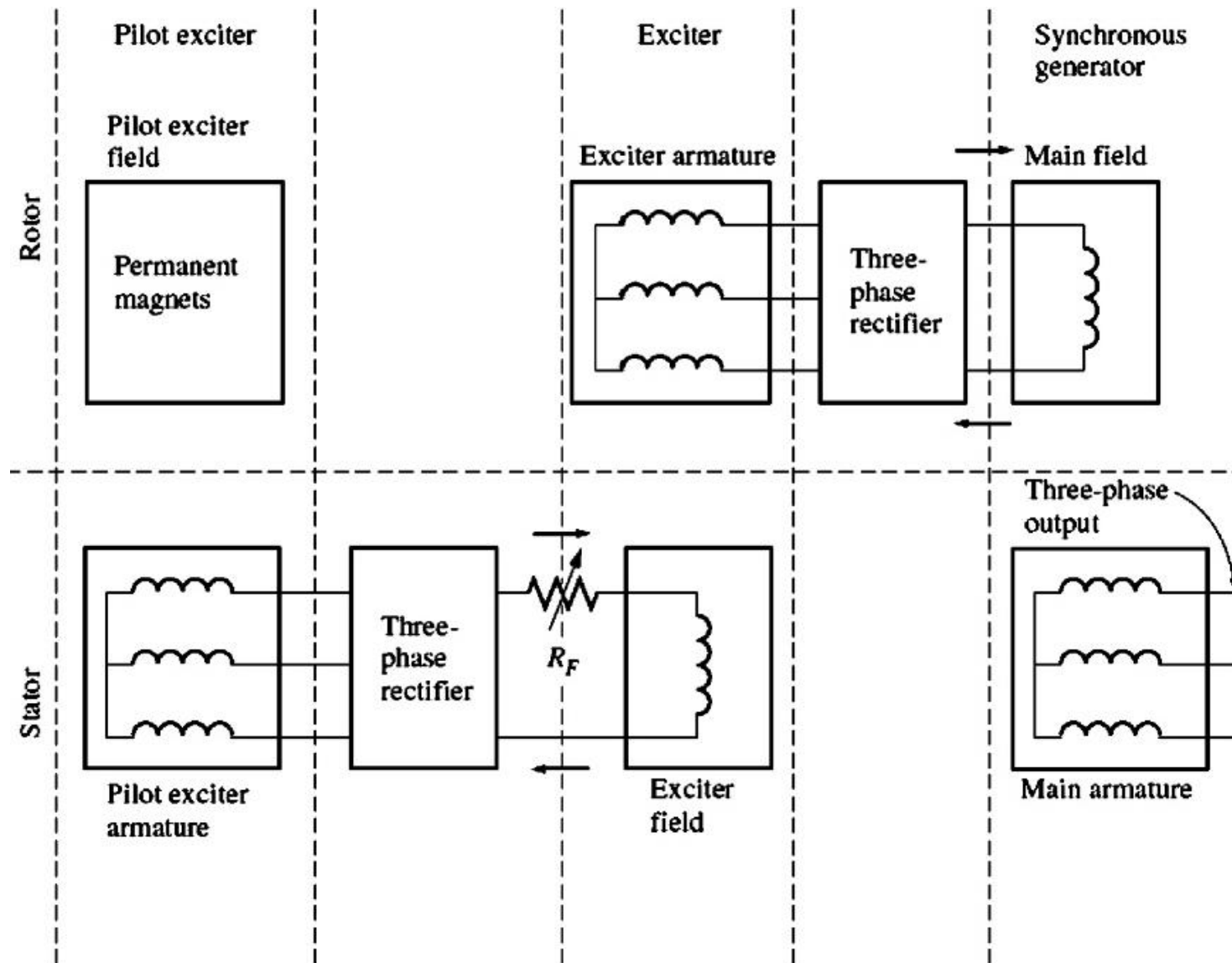
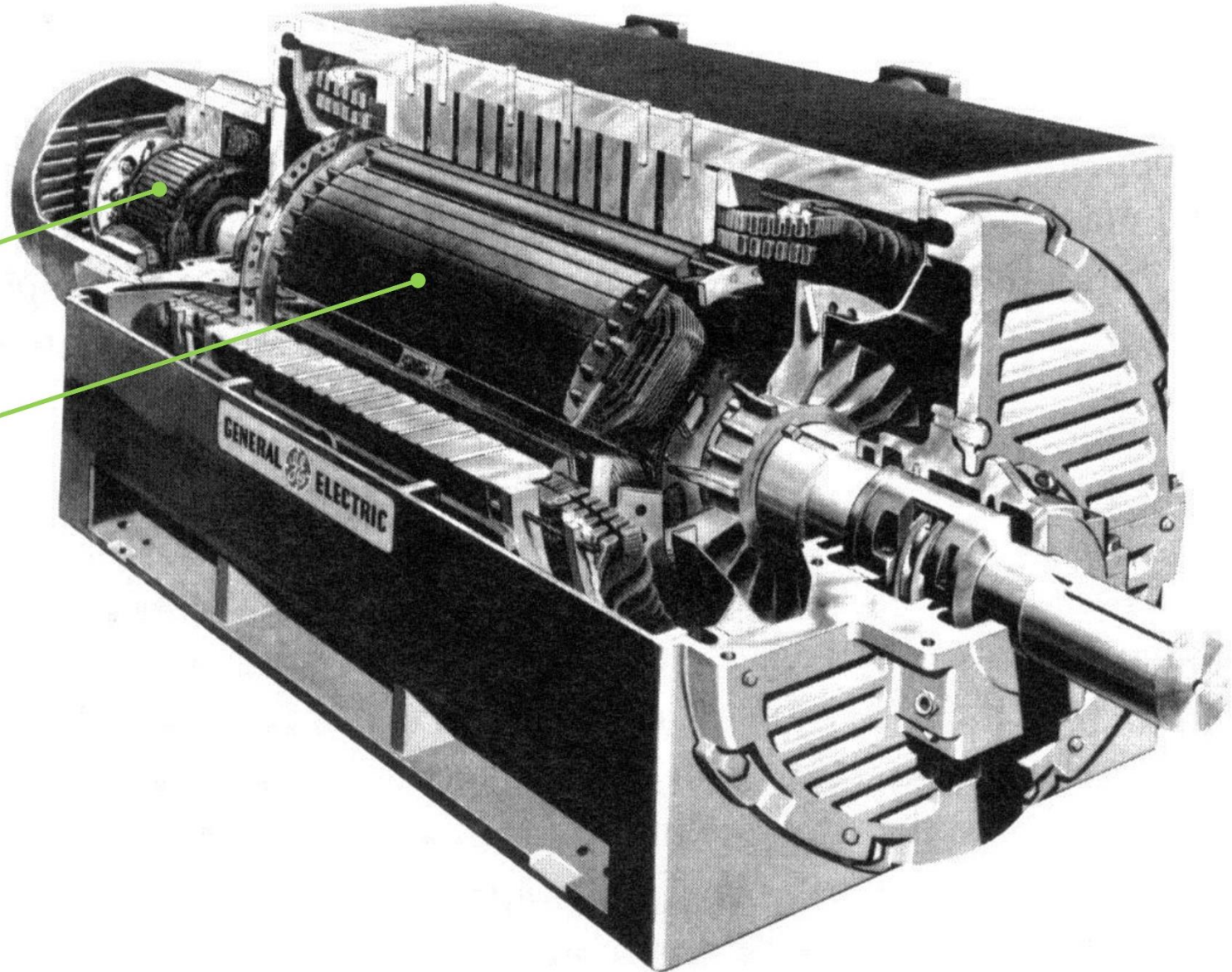


FIGURE 5-5

A brushless excitation scheme that includes a pilot exciter. The permanent magnets of the pilot exciter produce the field current of the exciter, which in turn produces the field current of the main machine.

Exciter

Salient poles.



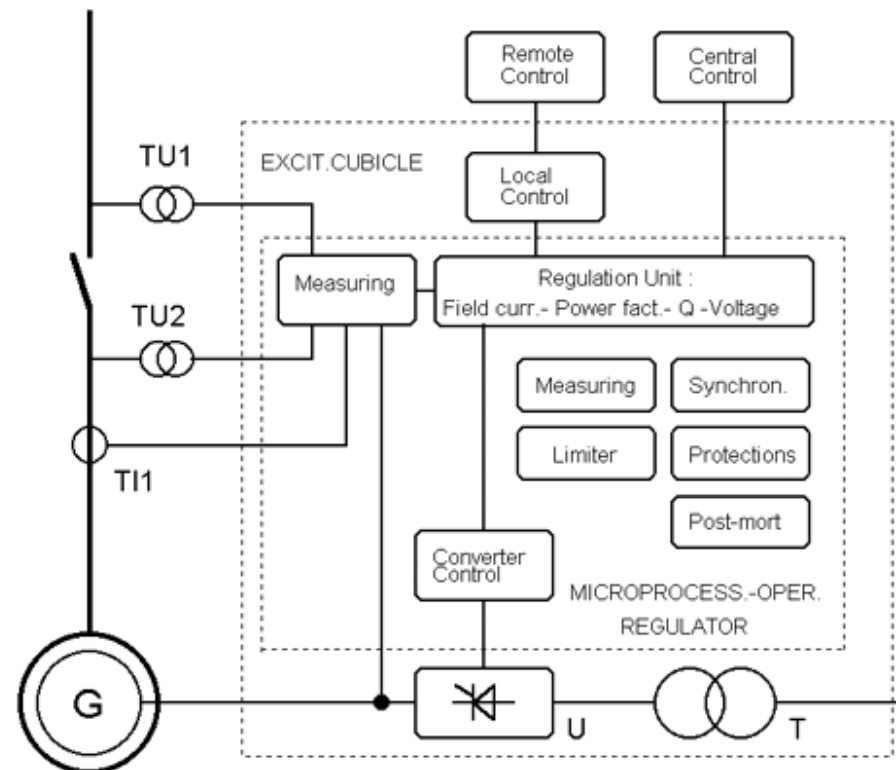
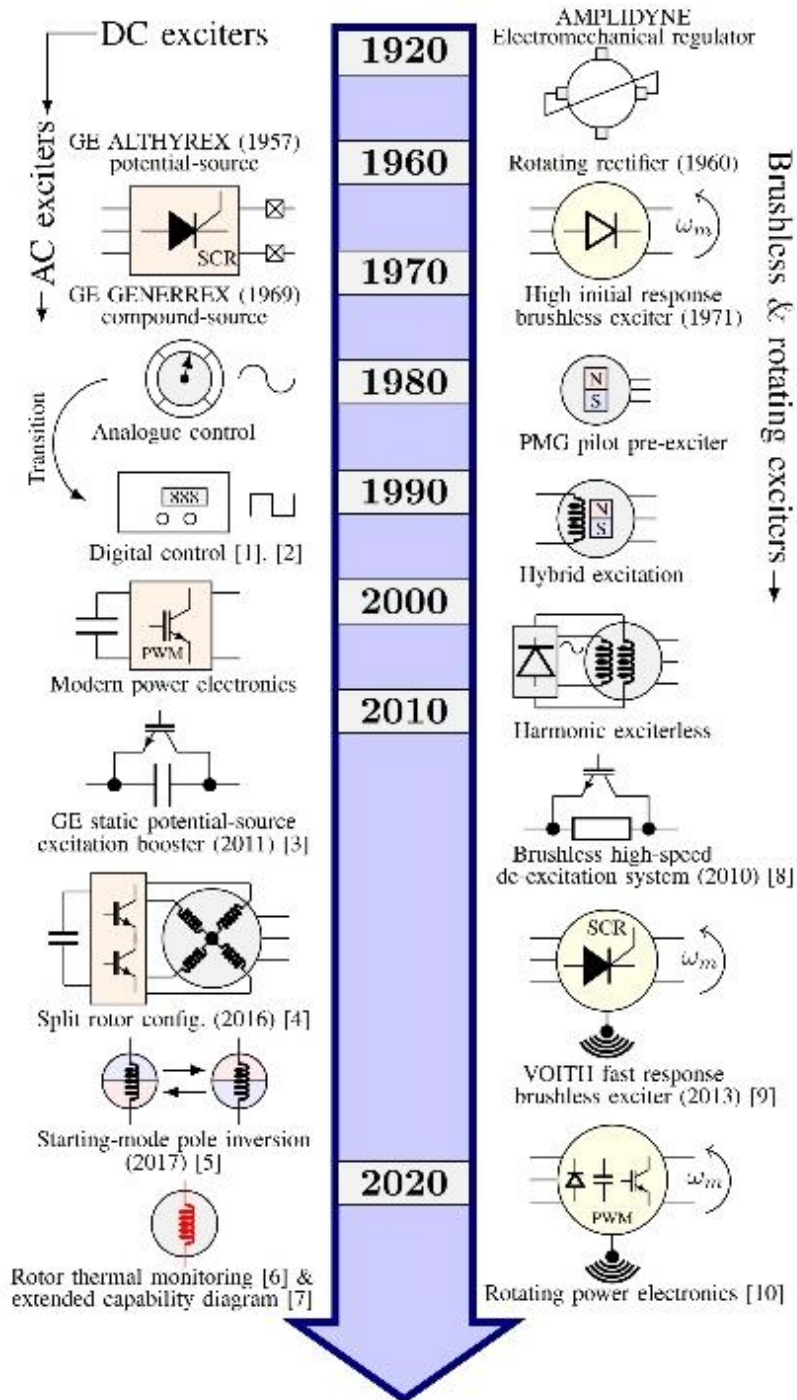
Present day Excitation systems (FYI)

3 topologies:

1. Static excitation
2. Rotating brushless excitation
3. Harmonic excited synchronous machines

Ref [4]: Continuous Evolution and Modern Approaches of Excitation Systems for Synchronous Machines

J. K. Nøland, Member, IEEE, M. Giset, Student Member, IEEE, and E. F. Alves, Member, IEEE



Example of static excitation system



Ref [4]:
<https://www.tenel.cz/en/static-excitation-systems-for-generators-and-motors.htm>

The speed of rotation of a SG

$$f_e = \frac{p}{120} n_m$$

Where f_e is the electrical frequency, Hz;
 n_m is the rotor speed of the machine,
 p is the number of poles.

$$\theta_e = \frac{P}{2} \theta_m$$

$$f_e = \frac{P}{2} f_m$$

$$\omega_e = \frac{P}{2} \omega_m$$

$$f_m = n_m/60.$$

Induced voltage in ac machines (4.4)

Basic theorem needed:

1. The equation for the force on a wire in the presence of a magnetic field:

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B}) \quad (1-43)$$

where $\mathbf{F}$ = force on wire
 i = magnitude of current in wire
 $\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow
 $\mathbf{B}$ = magnetic flux density vector

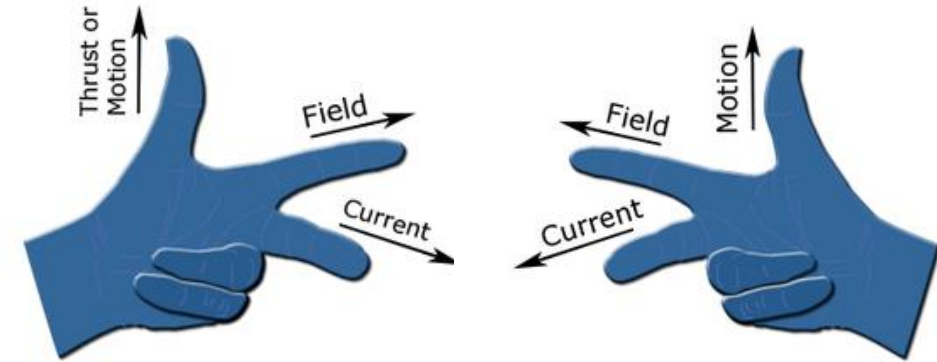
2. The equation for the voltage induced on a wire moving in a magnetic field:

$$e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \quad (1-45)$$

where e_{ind} = voltage induced in wire
 $\mathbf{v}$ = velocity of the wire
 $\mathbf{B}$ = magnetic flux density vector
 $\mathbf{l}$ = length of conductor in the magnetic field

Things to know:

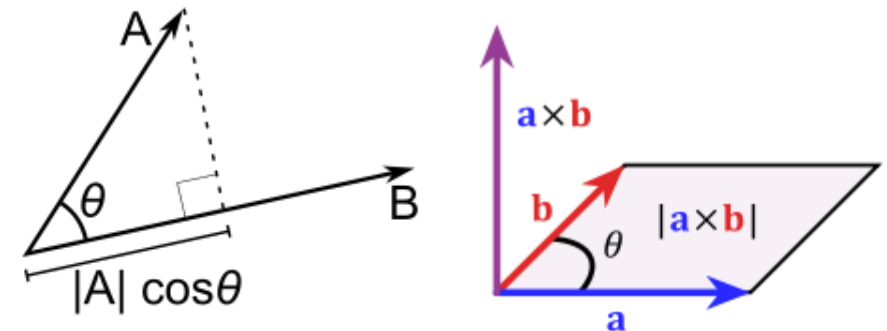
1. Fleming's left and right hand rule

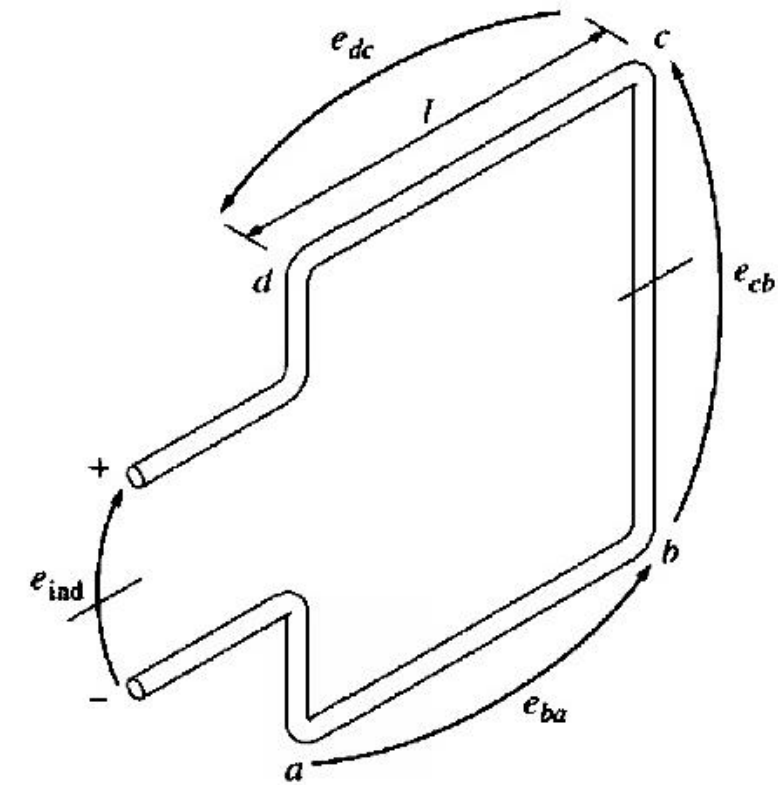


For Motor

For Generator

2. Dot product and cross product of vectors



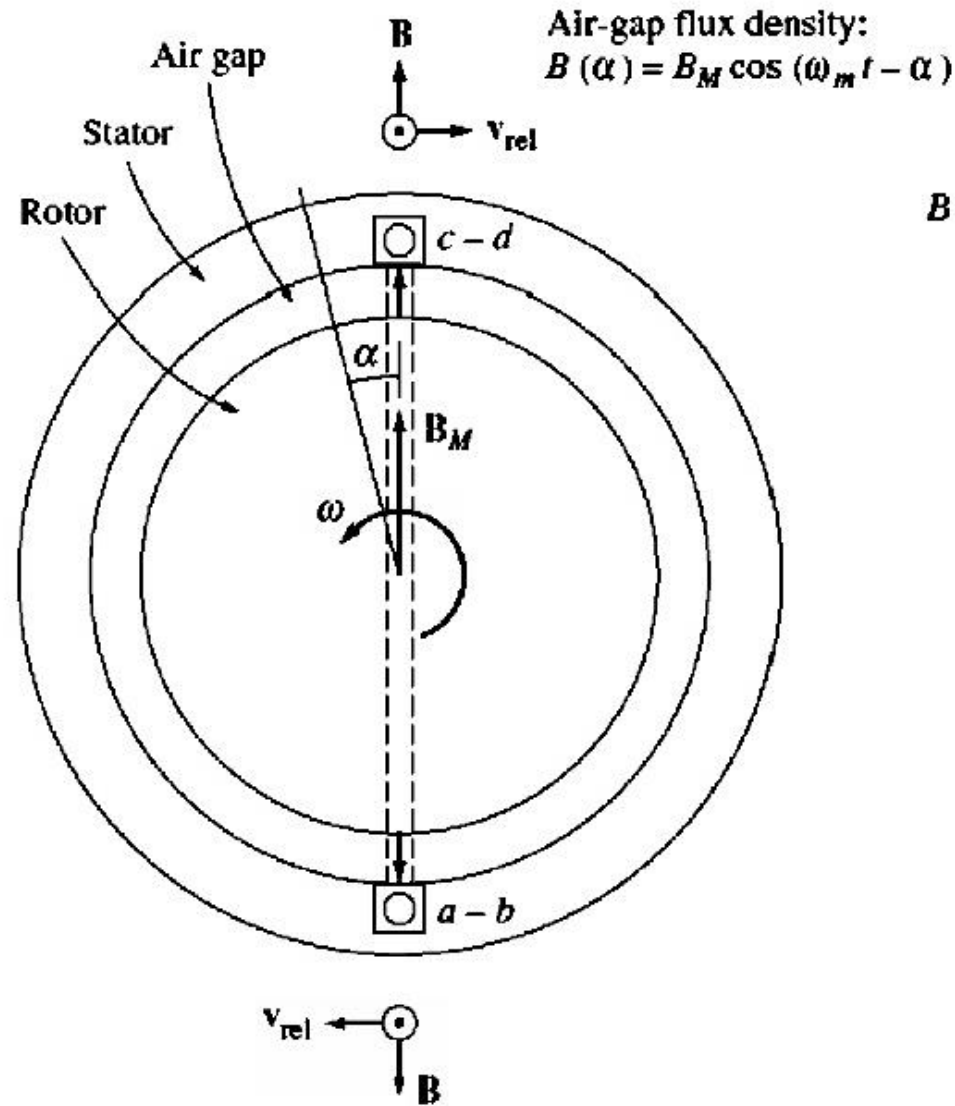


$$B = B_M \cos \alpha$$

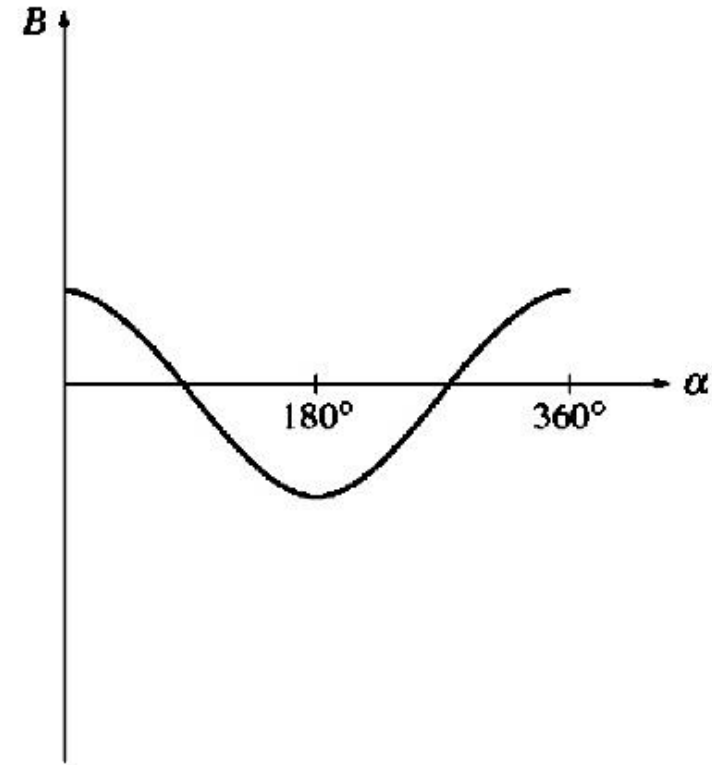
$$B = B_M \cos(\omega t - \alpha)$$

The equation for the induced voltage in a wire is

$$e = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$



Voltage is really into the page,
 since B is negative here.



(c)

Need to study also: 4.1 (FYI)

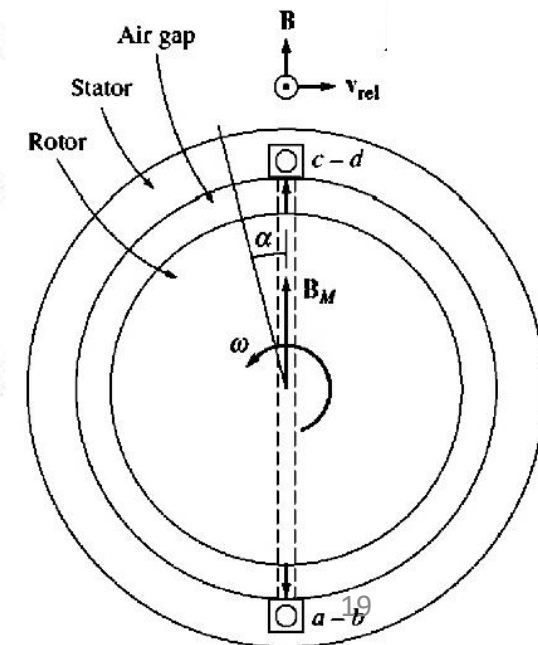
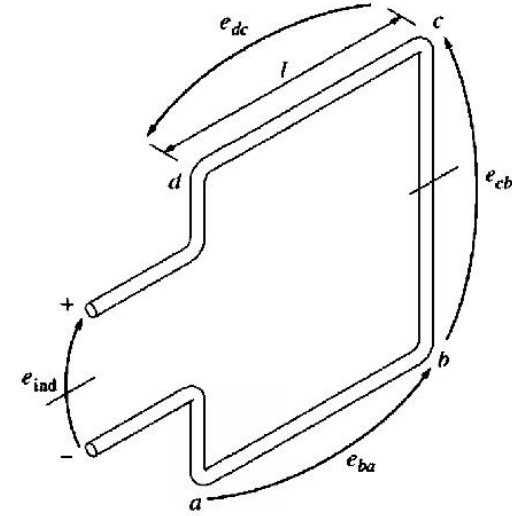
1. *Segment ab.* For segment *ab*, $\alpha = 180^\circ$. Assuming that $\mathbf{B}$ is directed radially outward from the rotor, the angle between $\mathbf{v}$ and $\mathbf{B}$ in segment *ab* is 90° , while the quantity $\mathbf{v} \times \mathbf{B}$ is in the direction of $\mathbf{l}$, so

$$\begin{aligned}
 e_{ba} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \\
 &= vBl \quad \text{directed out of the page} \\
 &= -v[B_M \cos(\omega_m t - 180^\circ)]l \\
 &= -vB_M l \cos(\omega_m t - 180^\circ)
 \end{aligned} \tag{4-38}$$

where the minus sign comes from the fact that the voltage is built up with a polarity opposite to the assumed polarity.

2. *Segment bc.* The voltage on segment *bc* is zero, since the vector quantity $\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{l}$, so

$$e_{cb} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = 0 \tag{4-39}$$



3. Segment cd . For segment cd , the angle $\alpha = 0^\circ$. Assuming that $\mathbf{B}$ is directed radially outward from the rotor, the angle between $\mathbf{v}$ and $\mathbf{B}$ in segment cd is 90° , while the quantity $\mathbf{v} \times \mathbf{B}$ is in the direction of $\mathbf{l}$, so

$$\begin{aligned}
 e_{dc} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \\
 &= vBl \quad \text{directed out of the page} \\
 &= v(B_M \cos \omega_m t)l \\
 &= vB_M l \cos \omega_m t
 \end{aligned} \tag{4-40}$$

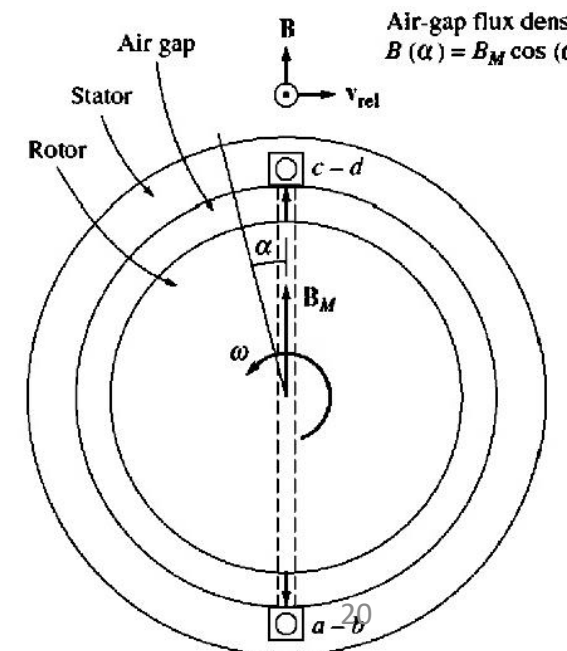
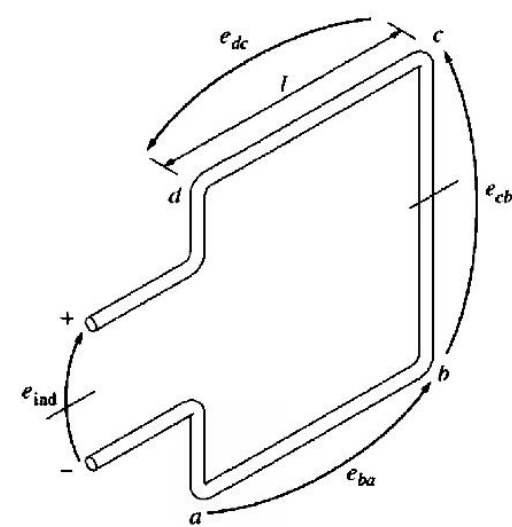
4. Segment da . The voltage on segment da is zero, since the vector quantity $\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{l}$, so

$$e_{ad} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = 0 \tag{4-41}$$

Therefore, the total voltage on the coil will be

$$\begin{aligned}
 e_{\text{ind}} &= e_{ba} + e_{dc} \\
 &= -vB_M l \cos(\omega_m t - 180^\circ) + vB_M l \cos \omega_m t
 \end{aligned} \tag{4-42}$$

Since $\cos \theta = -\cos (\theta - 180^\circ)$,



$$\begin{aligned}
 e_{\text{ind}} &= vB_M l \cos \omega_m t + vB_M l \cos \omega_m t \\
 &= 2vB_M l \cos \omega_m t
 \end{aligned}
 \tag{4-43}$$

Since the velocity of the end conductors is given by $v = r\omega_m$, Equation (4-43) can be rewritten as

$$\begin{aligned}
 e_{\text{ind}} &= 2(r\omega_m)B_M l \cos \omega_m t \\
 &= 2rlB_M \omega_m \cos \omega_m t
 \end{aligned}$$

Finally, the flux passing through the coil can be expressed as $\phi = 2rlB_m$ (see Problem 4-7), while $\omega_m = \omega_e = \omega$ for a two-pole stator, so the induced voltage can be expressed as

$$\boxed{e_{\text{ind}} = \phi \omega \cos \omega t}
 \tag{4-44}$$

Equation (4-44) describes the voltage induced in a single-turn coil. If the coil in the stator has N_C turns of wire, then the total induced voltage of the coil will be

$$\boxed{e_{\text{ind}} = N_C \phi \omega \cos \omega t}
 \tag{4-45}$$

The Induced Voltage in a Three-Phase Set of Coils

$$e_{aa'}(t) = N_C \phi \omega \sin \omega t \quad \text{V}$$

$$e_{bb'}(t) = N_C \phi \omega \sin (\omega t - 120^\circ) \quad \text{V}$$

$$e_{cc'}(t) = N_C \phi \omega \sin (\omega t - 240^\circ) \quad \text{V}$$

The RMS Voltage in a Three-Phase Stator

The peak voltage in any phase of a three-phase stator of this sort is

$$E_{\max} = N_C \phi \omega \quad (4-47)$$

Since $\omega = 2\pi f$, this equation can also be written as

$$E_{\max} = 2\pi N_C \phi f \quad (4-48)$$

Therefore, the rms voltage of any phase of this three-phase stator is

$$E_A = \frac{2\pi}{\sqrt{2}} N_C \phi f \quad (4-49)$$

$$\boxed{E_A = \sqrt{2}\pi N_C \phi f} \quad (4-50)$$

The internal generated voltage of a synchronous generator (5.3)

$$\boxed{E_A = K\phi\omega} \quad (5-1)$$

where K is a constant representing the construction of the machine. If ω is expressed in *electrical* radians per second, then

$$K = \frac{N_c}{\sqrt{2}} \quad (5-2)$$

while if ω is expressed in *mechanical* radians per second, then

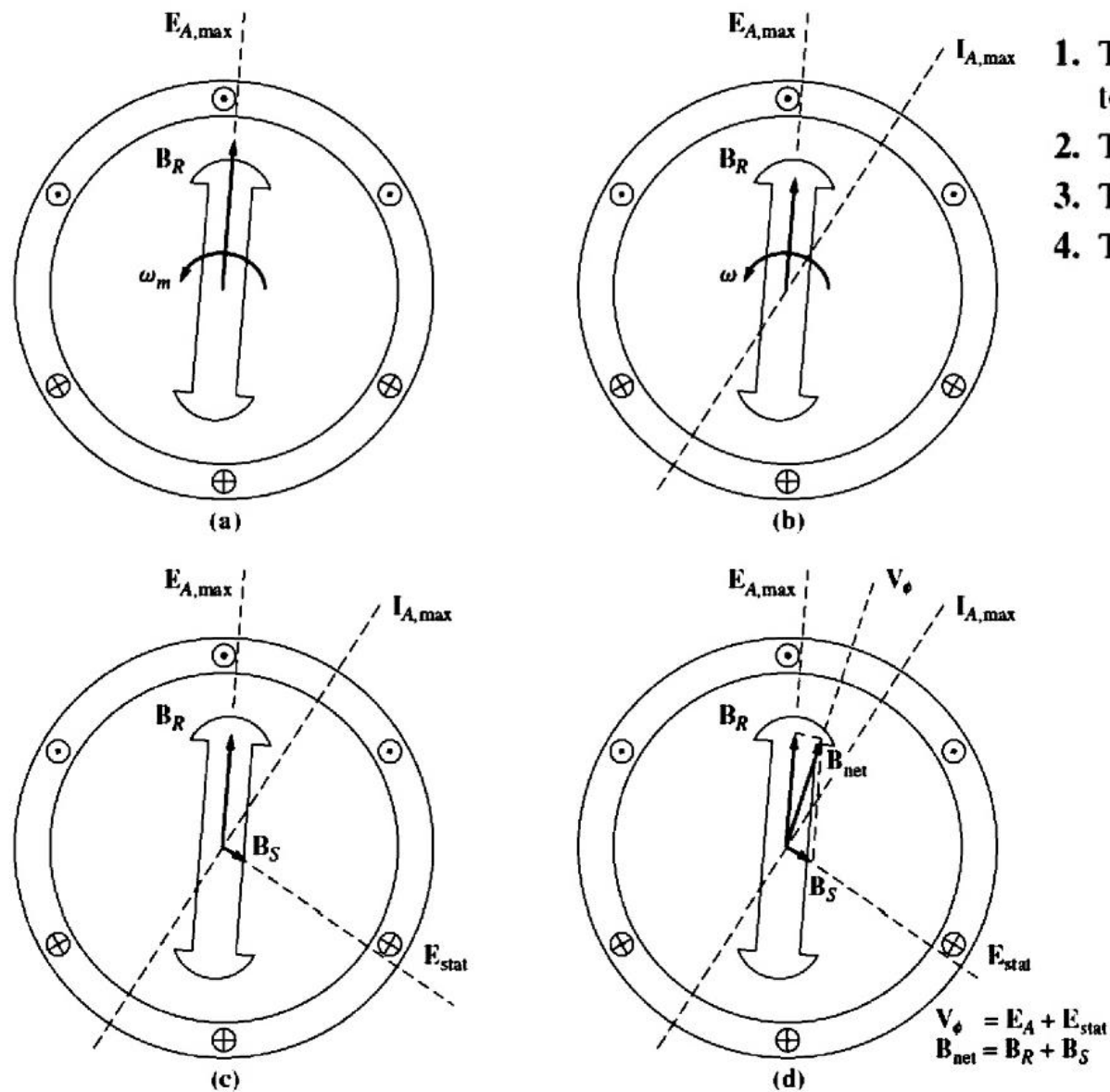
$$K = \frac{N_c P}{\sqrt{2}} \quad (5-3)$$

Equivalent circuit of a SG (5.4)

There are a number of factors that cause the difference between E_A and V_ϕ :

1. The distortion of the air-gap magnetic field by the current flowing in the stator, called *armature reaction*.
2. The self-inductance of the armature coils.
3. The resistance of the armature coils.
4. The effect of salient-pole rotor shapes.

- When the rotor of a synchronous generator is spinning, a voltage E_A is induced in its stator.
- When a load is connected, a current starts flowing creating a magnetic field in machine's stator.
- This stator magnetic field B_S adds to the rotor (main) magnetic field B_R affecting the total magnetic field and, therefore, the phase voltage.

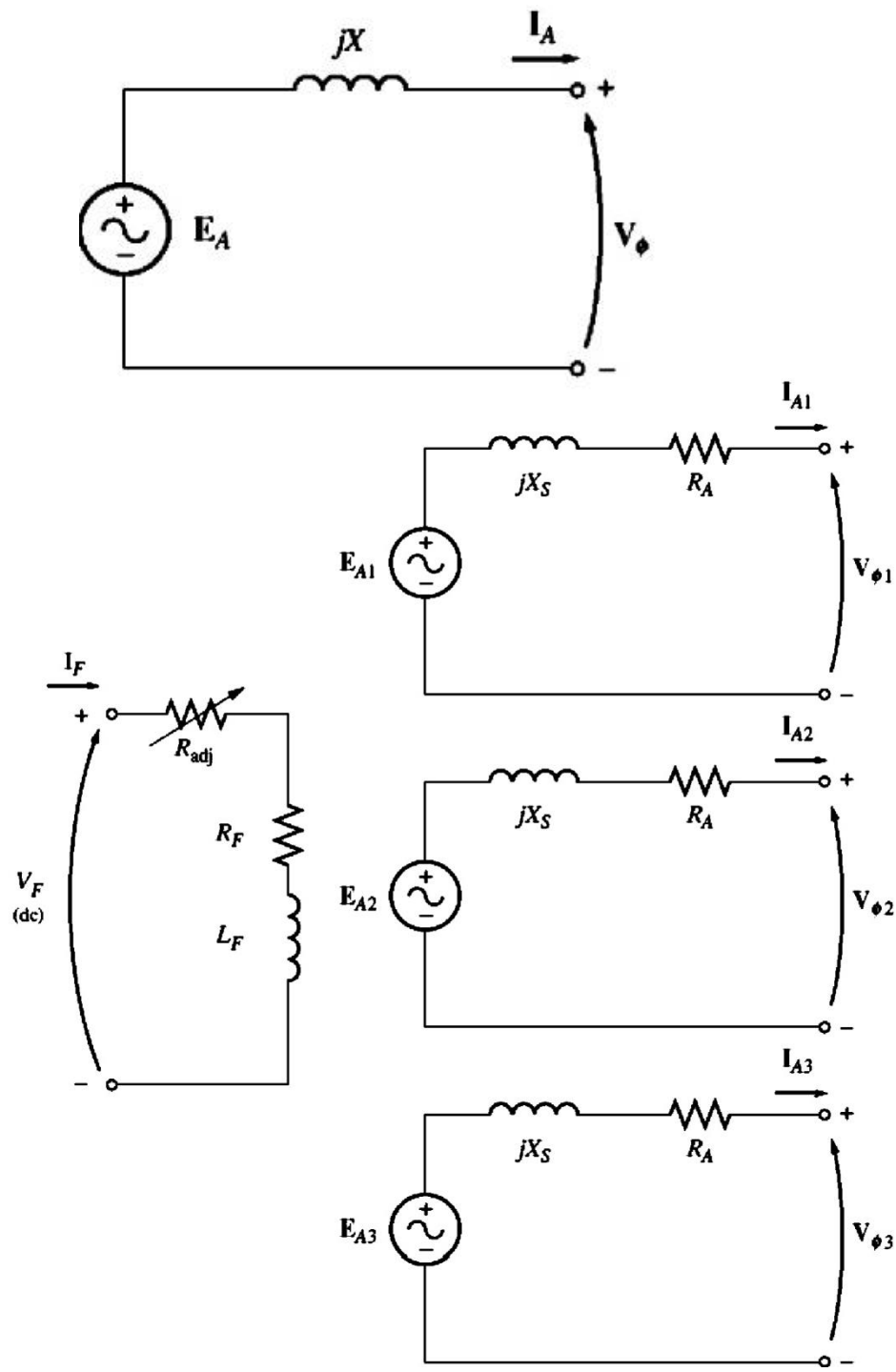


With two voltages present in the stator windings, the total voltage in a phase is just the *sum* of the internal generated voltage E_A and the armature reaction voltage E_{stat} :

$$V_\phi = E_A + E_{stat} \quad (5-4)$$

The net magnetic field B_{net} is just the sum of the rotor and stator magnetic fields:

$$B_{net} = B_R + B_S \quad (5-5)$$



Since the angles of E_A and B_R are the same and the angles of E_{stat} and B_S are the same, the resulting magnetic field B_{net} will coincide with the net voltage V_ϕ .

If X is a constant of proportionality, then *the armature reaction voltage can be expressed as*

$$E_{stat} = -jXI_A \quad (5-6)$$

The voltage on a phase is thus

$$\boxed{V_\phi = E_A - jXI_A} \quad (5-7)$$

In addition to the effects of armature reaction, the stator coils have a self-inductance and a resistance. If the stator self-inductance is called L_A (and its corresponding reactance is called X_A) while the stator resistance is called R_A , then the total difference between E_A and V_ϕ is given by

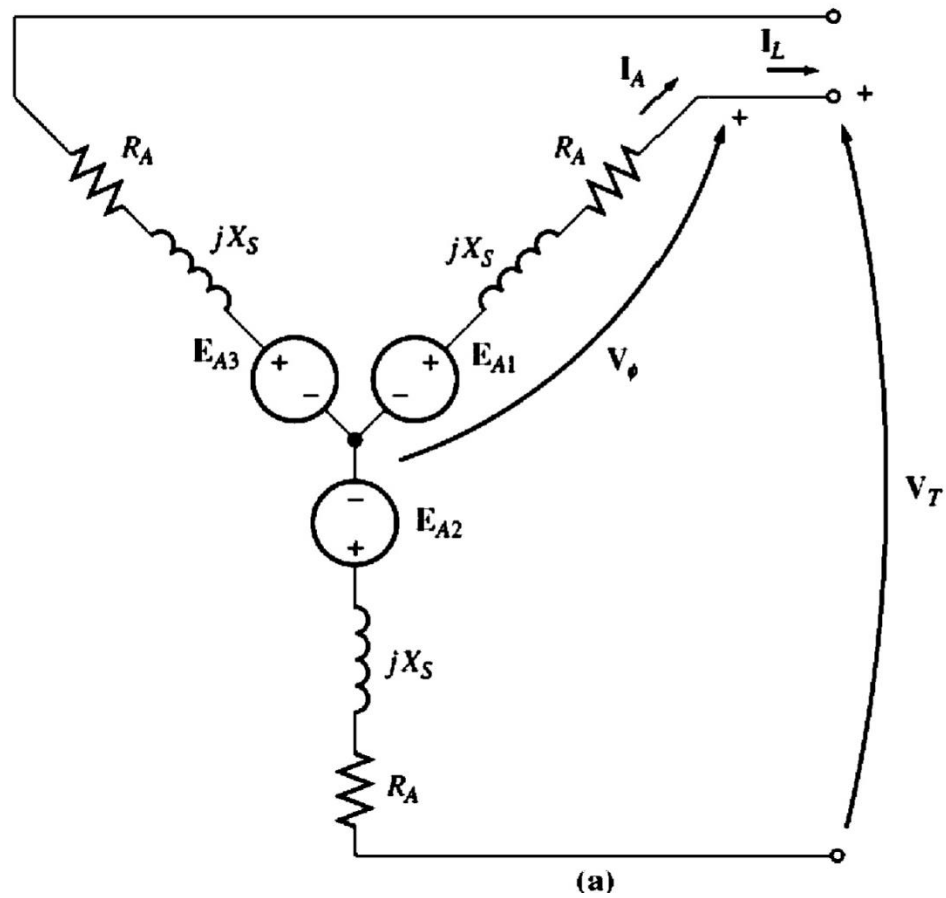
$$V_\phi = E_A - jXI_A - jX_AI_A - R_AI_A \quad (5-9)$$

The armature reaction effects and the self-inductance in the machine are both represented by reactances, and it is customary to combine them into a single reactance, called the *synchronous reactance* of the machine:

$$X_S = X + X_A \quad (5-10)$$

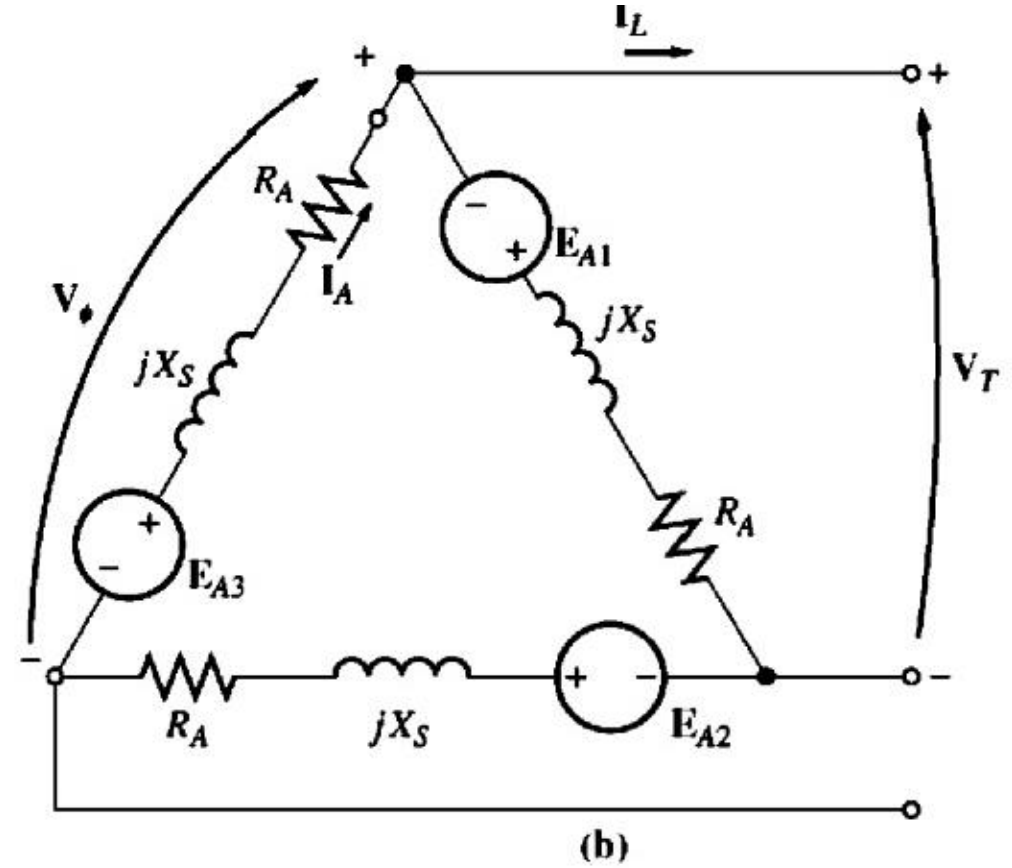
Therefore, the final equation describing V_ϕ is

$$\boxed{V_\phi = E_A - jX_S I_A - R_A I_A} \quad (5-11)$$



If they are Y-connected, then the terminal voltage V_T is related to the phase voltage by

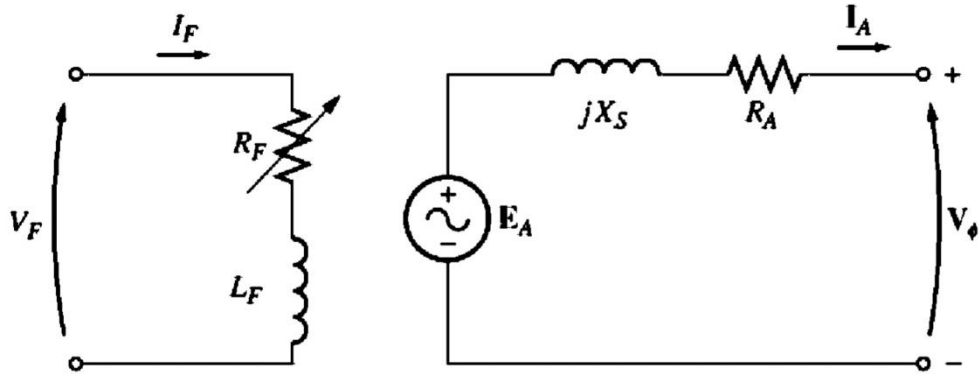
$$V_T = \sqrt{3}V_\phi \quad (5-12)$$



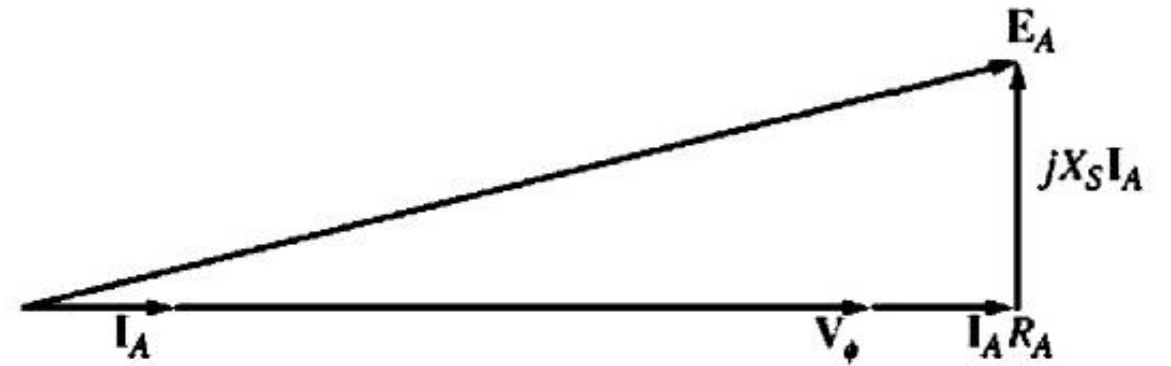
If they are Δ -connected, then

$$V_T = V_\phi \quad (5-13)$$

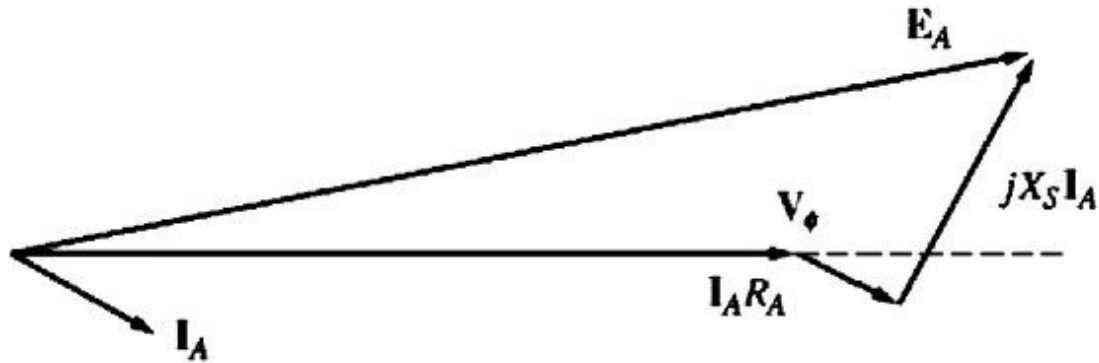
Phasor diagram of a sg (5.5)



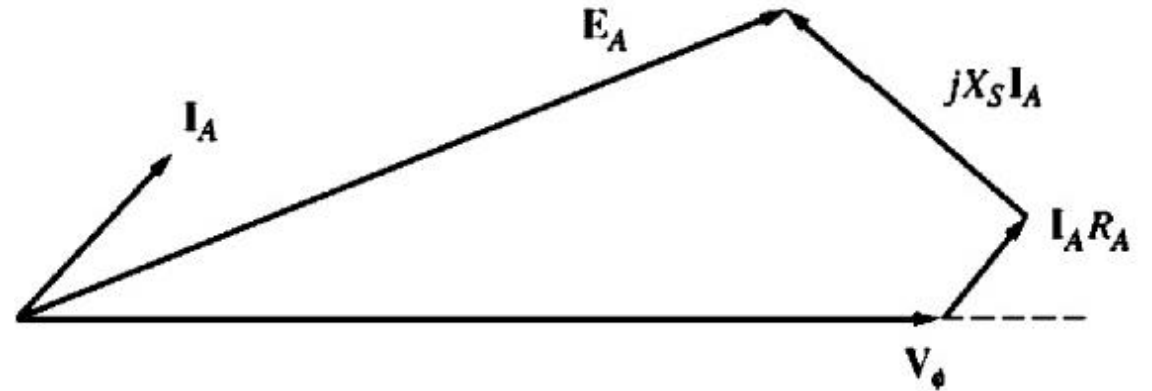
$$\mathbf{V}_\phi = \mathbf{E}_A - j\mathbf{X}_S \mathbf{I}_A - \mathbf{R}_A \mathbf{I}_A$$



For Resistive Load

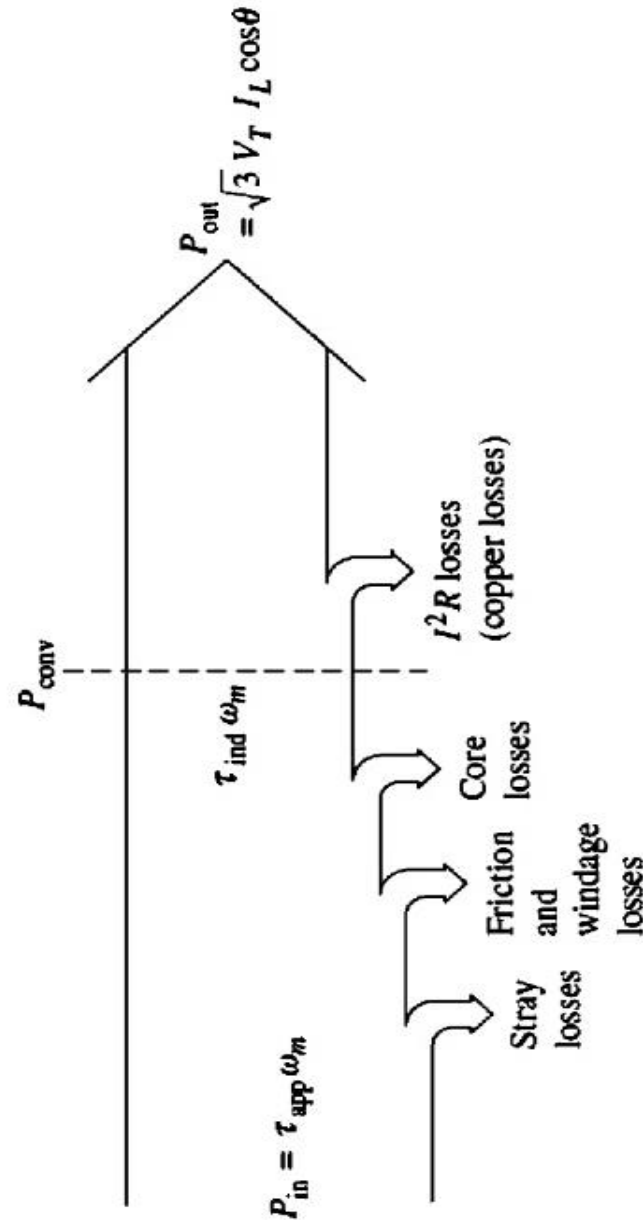


For Inductive Load



For Capacitive Load

Power and torque in SG (5.6)



The input mechanical power is the shaft power in the generator $P_{in} = \tau_{app} \omega_m$, while the power converted from mechanical to electrical form internally is given by

$$P_{conv} = \tau_{ind} \omega_m \quad (5-14)$$

$$= 3E_A I_A \cos \gamma \quad (5-15)$$

where γ is the angle between E_A and I_A . The difference between the input power to the generator and the power converted in the generator represents the mechanical, core, and stray losses of the machine.

The real electrical output power of the synchronous generator can be expressed in line quantities as

$$P_{out} = \sqrt{3} V_T I_L \cos \theta \quad (5-16)$$

and in phase quantities as

$$P_{out} = 3 V_\phi I_A \cos \theta \quad (5-17)$$

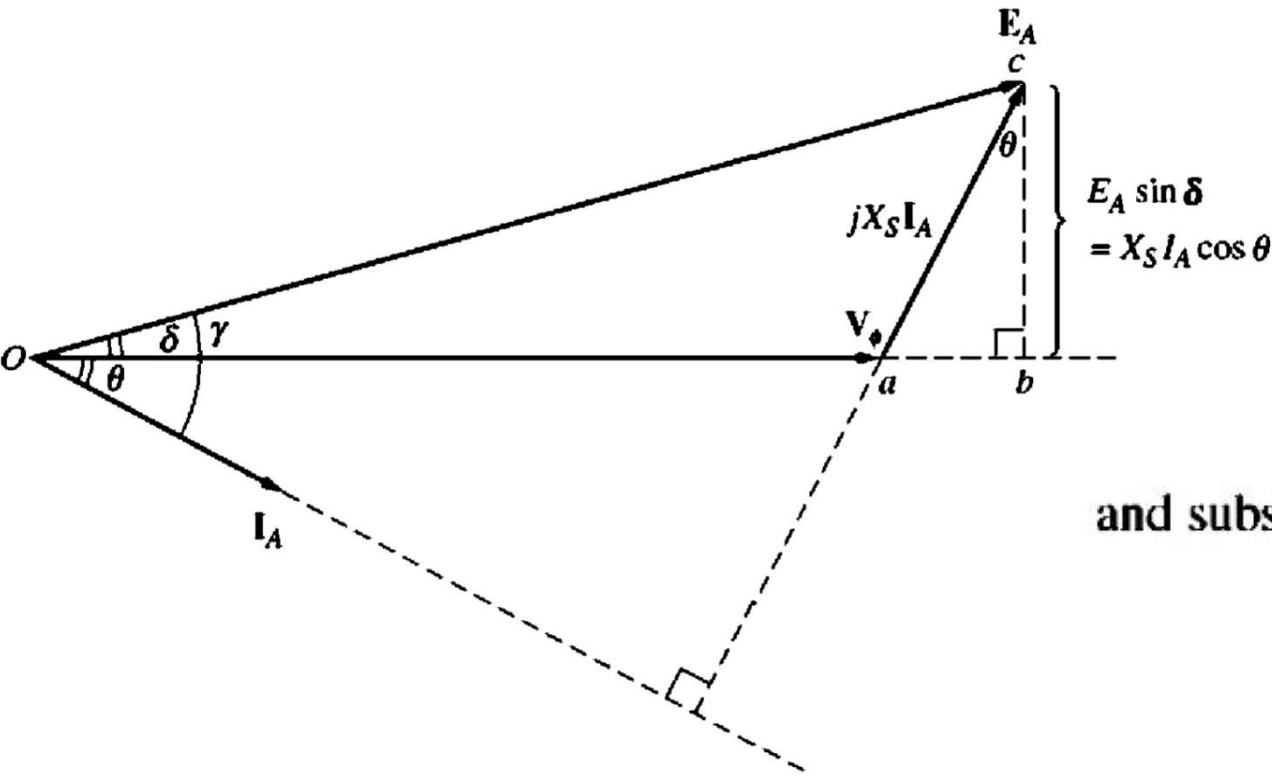
The reactive power output can be expressed in line quantities as

$$Q_{out} = \sqrt{3} V_T I_L \sin \theta \quad (5-18)$$

or in phase quantities as

$$Q_{out} = 3 V_\phi I_A \sin \theta \quad (5-19)$$

Since, $X_s \gg R_a$



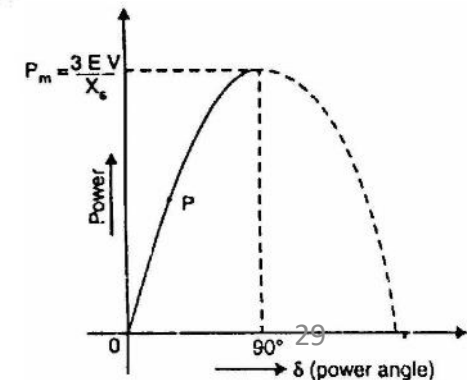
$$I_A \cos \theta = \frac{E_A \sin \delta}{X_s}$$

and substituting this expression into Equation (5-17) gives

$$P = \frac{3V_\phi E_A \sin \delta}{X_s} \quad (5-20)$$

Equation (5-20) shows that the power produced by a synchronous generator depends on the angle δ between V_ϕ and E_A . The angle δ is known as the *torque angle* of the machine. Notice also that the maximum power that the generator can supply occurs when $\delta = 90^\circ$. At $\delta = 90^\circ$, $\sin \delta = 1$, and

$$P_{\max} = \frac{3V_\phi E_A}{X_s}$$



From Chapter 4, the induced torque in this generator can be expressed as

$$\tau_{\text{ind}} = k\mathbf{B}_R \times \mathbf{B}_S \quad (4-58)$$

or as

$$\tau_{\text{ind}} = k\mathbf{B}_R \times \mathbf{B}_{\text{net}} \quad (4-60)$$

Derivation of torque equation – self study

The magnitude of Equation (4-60) can be expressed as

$$\tau_{\text{ind}} = kB_R B_{\text{net}} \sin \delta \quad (4-61)$$

where δ is the angle between the rotor and net magnetic fields (the so-called *torque angle*). Since $\mathbf{B}_R$ produces the voltage $\mathbf{E}_A$ and $\mathbf{B}_{\text{net}}$ produces the voltage $\mathbf{V}_\phi$, the angle δ between $\mathbf{E}_A$ and $\mathbf{V}_\phi$ is the same as the angle δ between $\mathbf{B}_R$ and $\mathbf{B}_{\text{net}}$.

An alternative expression for the induced torque in a synchronous generator can be derived from Equation (5-20). Because $P_{\text{conv}} = \tau_{\text{ind}}\omega_m$, the induced torque can be expressed as

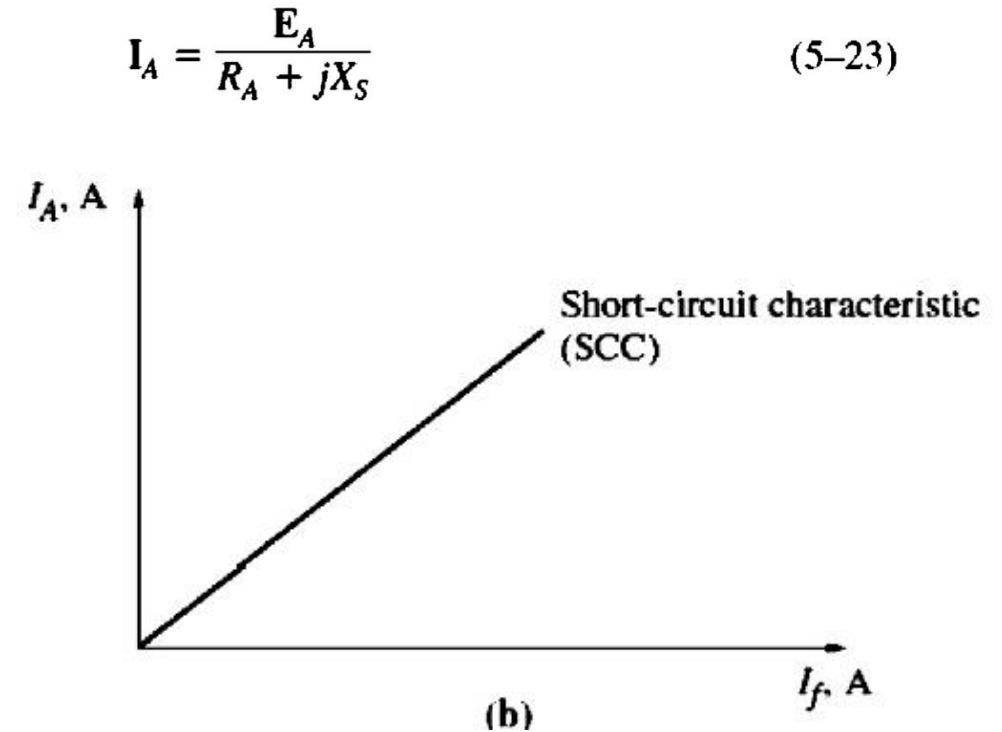
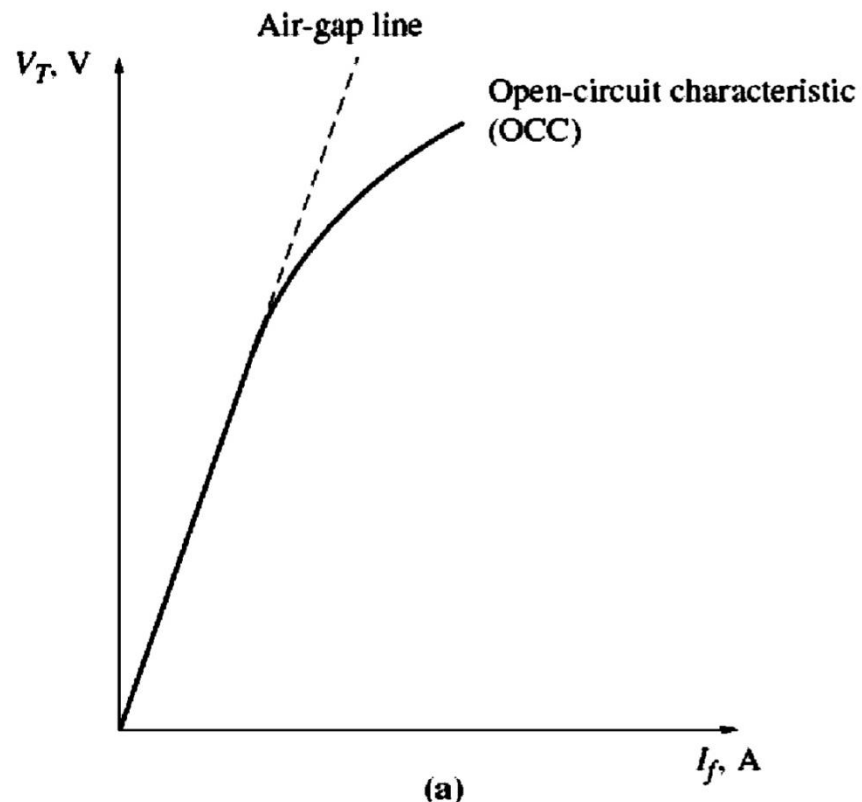
$$\tau_{\text{ind}} = \frac{3V_\phi E_A \sin \delta}{\omega_m X_S} \quad (5-22)$$

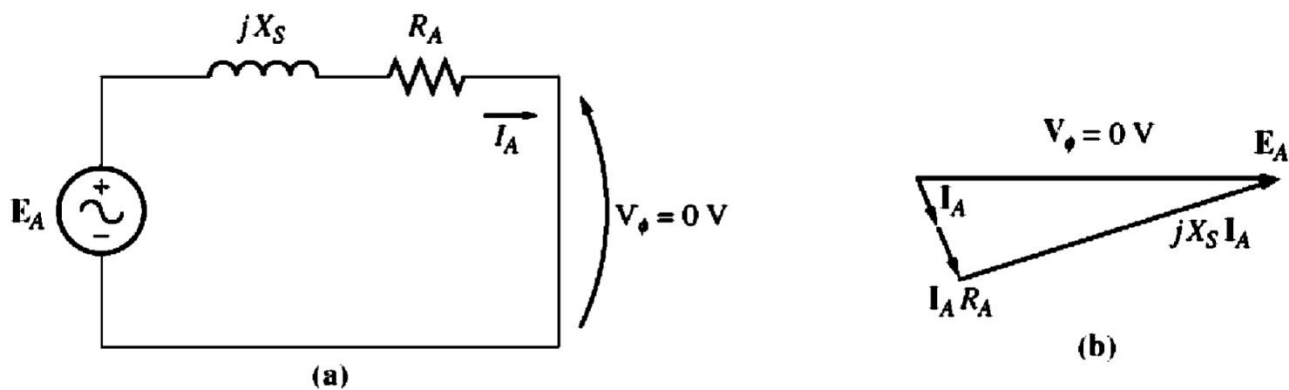
This expression describes the induced torque in terms of electrical quantities, whereas Equation (4-60) gives the same information in terms of magnetic quantities.

Measuring SG model parameters (5.7)

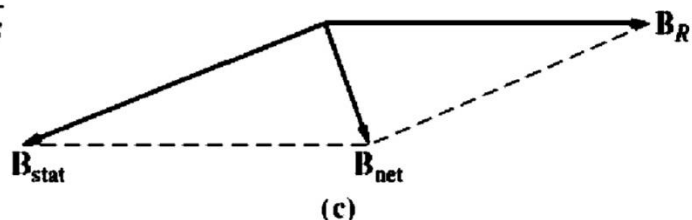
Quantities needed to be determined:

1. The relationship between field current and flux (and therefore between the field current and E_A)
2. The synchronous reactance
3. The armature resistance





$$I_A = \frac{E_A}{R_A + jX_S}$$



$$I_A = \frac{E_A}{\sqrt{R_A^2 + X_S^2}}$$

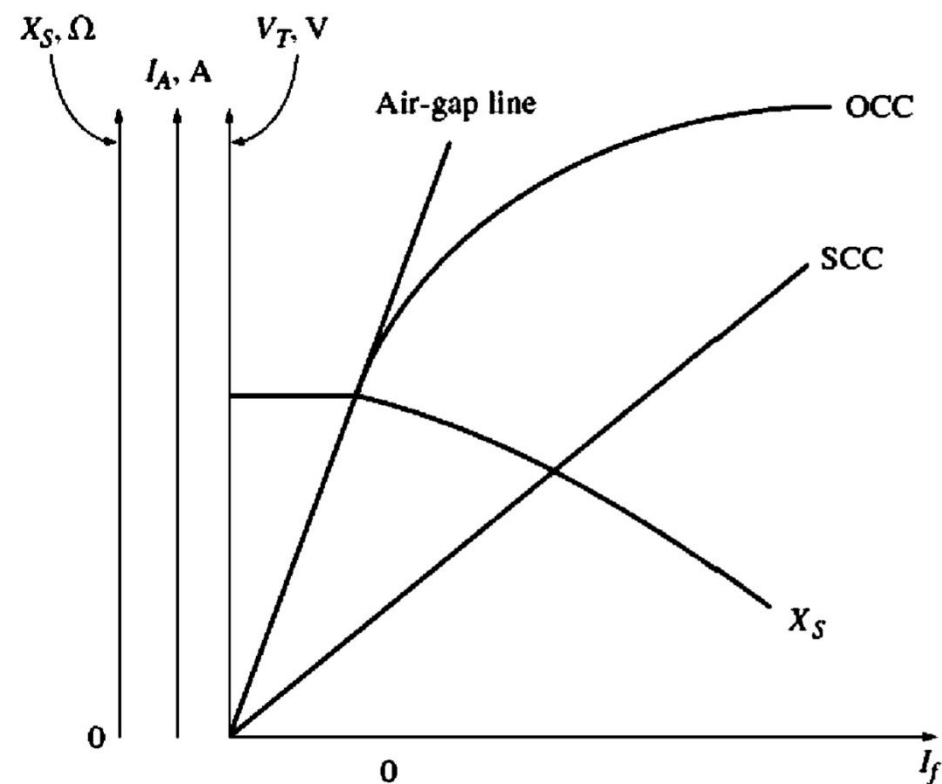
To understand what information these two characteristics yield, notice that, with V_ϕ equal to zero in Figure 5-18, the *internal machine impedance* is given by

$$Z_S = \sqrt{R_A^2 + X_S^2} = \frac{E_A}{I_A} \quad (5-25)$$

Since $X_S \gg R_A$, this equation reduces to

$$X_S \approx \frac{E_A}{I_A} = \frac{V_{\phi,oc}}{I_A} \quad (5-26)$$

If E_A and I_A are known for a given situation, then the synchronous reactance X_S can be found.



Problem with this method:

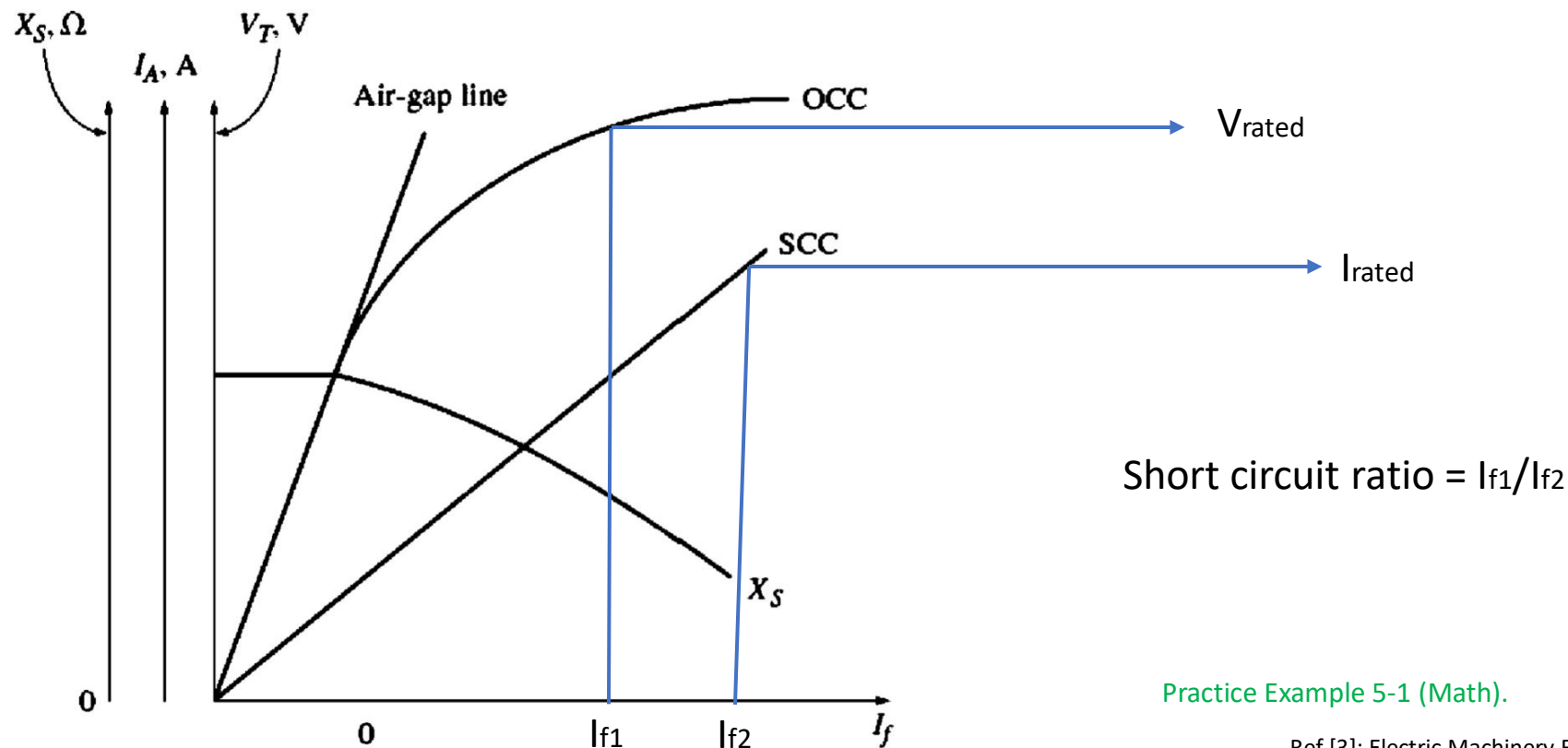


There is a problem with this approach, however. The internal generated voltage E_A comes from the OCC, where the machine is partially *saturated* for large field currents, while I_A is taken from the SCC, where the machine is *unsaturated* at all field currents. Therefore, at higher field currents, the E_A taken from the OCC at a given field current is *not* the same as the E_A at the same field current under short-circuit conditions, and this difference makes the resulting value of X_s only approximate.

However, the answer given by this approach is accurate up to the point of saturation, so the *unsaturated synchronous reactance* $X_{s,u}$ of the machine can be found simply by applying Equation (5–26) at any field current in the linear portion (on the air-gap line) of the OCC curve.

The short circuit ratio:

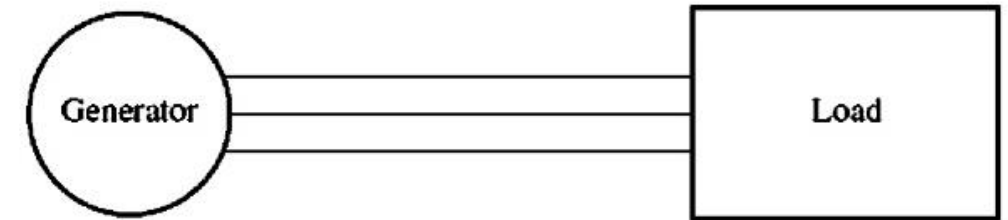
The *short-circuit ratio* of a generator is defined as the ratio of the *field current required for the rated voltage at open circuit* to the *field current required for the rated armature current at short circuit*. It can be shown that this quantity is just the reciprocal of the per-unit value of the approximate saturated synchronous reactance calculated by Equation (5-26).



The sg operating alone (5.8)

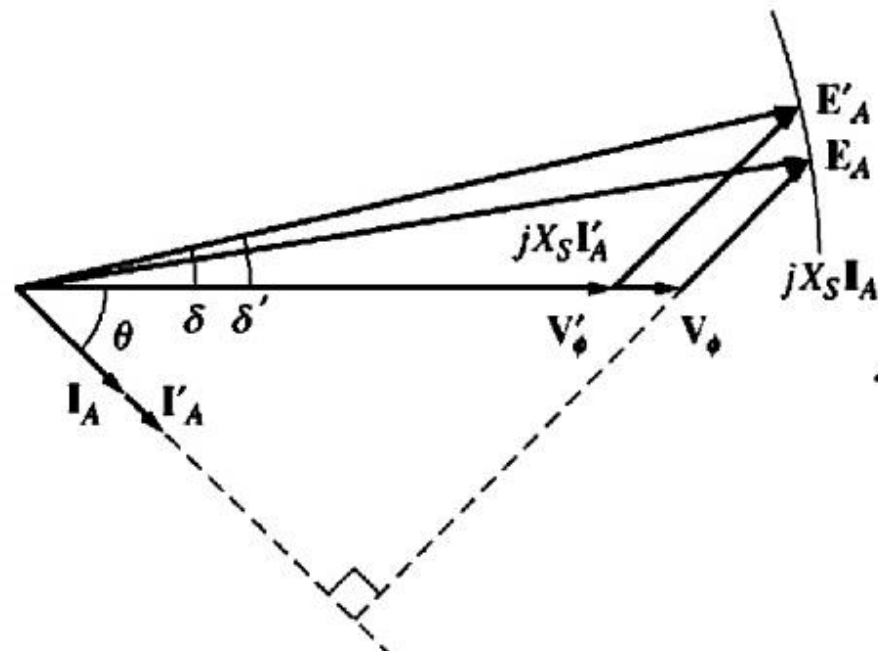
Assumptions:

1. Armature resistance, R_a is negligible
2. Speed of the generator is constant
3. The rotor flux is constant

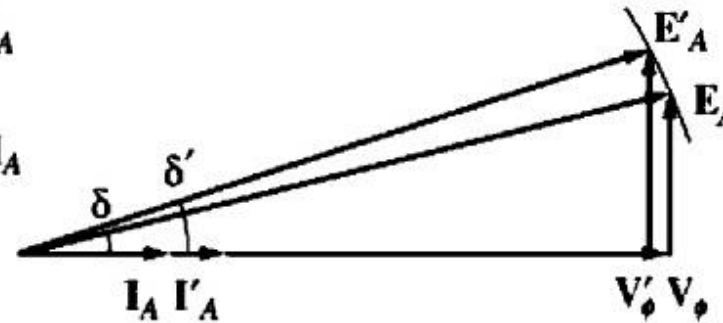


The effect of Load change on a synchronous generator operating alone:

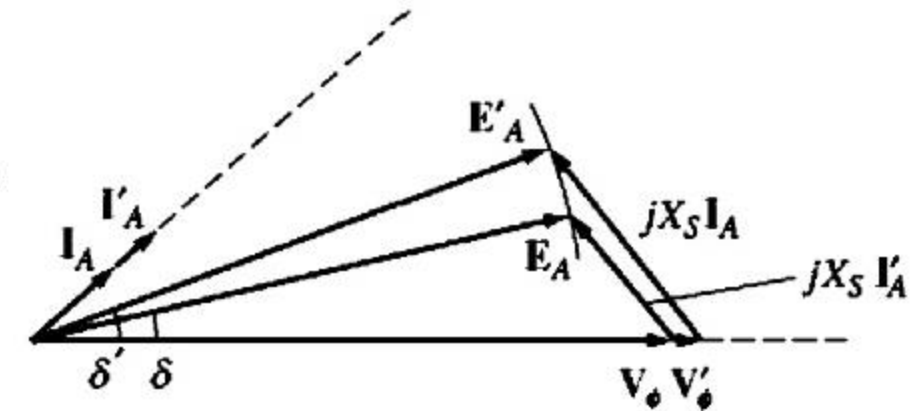
$$\mathbf{E}_A = \mathbf{V}_\phi + jX_S \mathbf{I}_A$$



For Inductive load



For Resistive load



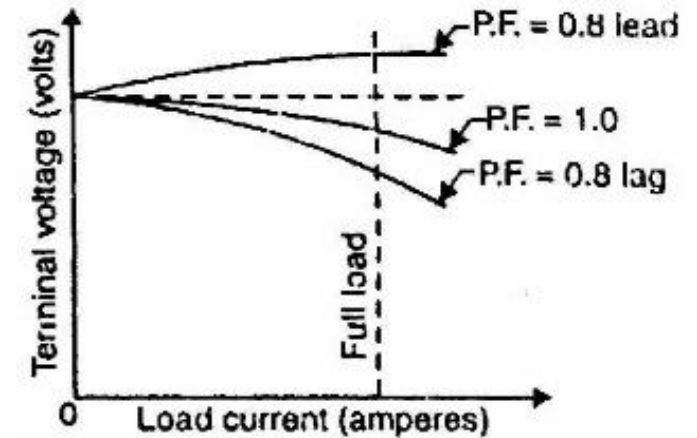
For Capacitive load



Torque angle (delta) is increasing with the increment of load

General conclusions from this discussion of synchronous generator behavior are

1. If lagging loads ($+Q$ or inductive reactive power loads) are added to a generator, V_ϕ and the terminal voltage V_T decrease significantly.
2. If unity-power-factor loads (no reactive power) are added to a generator, there is a slight decrease in V_ϕ and the terminal voltage.
3. If leading loads ($-Q$ or capacitive reactive power loads) are added to a generator, V_ϕ and the terminal voltage will rise.



Voltage regulation:

$$VR = \frac{V_{nl} - V_n}{V_n} \times 100\%$$

How to regulate terminal voltage (in case of lagging and unity load)

1. Decreasing the field resistance in the generator increases its field current.
2. An increase in the field current increases the flux in the machine.
3. An increase in the flux increases the internal generated voltage $E_A = K\phi\omega$.
4. An increase in E_A increases V_ϕ and the terminal voltage of the generator.

Math Practice sg1

Example 5–1. A 200-kVA, 480-V, 50-Hz, Y-connected synchronous generator with a rated field current of 5 A was tested, and the following data were taken:

1. $V_{T,OC}$ at the rated I_F was measured to be 540 V.
2. $I_{L,SC}$ at the rated I_F was found to be 300 A.
3. When a dc voltage of 10 V was applied to two of the terminals, a current of 25 A was measured.

Find the values of the armature resistance and the approximate synchronous reactance in ohms that would be used in the generator model at the rated conditions.

Solution:

The generator described above is Y-connected, so the direct current in the resistance test flows through two windings. Therefore, the resistance is given by

$$2R_A = \frac{V_{DC}}{I_{DC}}$$
$$R_A = \frac{V_{DC}}{2I_{DC}} = \frac{10 \text{ V}}{(2)(25 \text{ A})} = 0.2 \Omega$$

The internal generated voltage at the rated field current is equal to

$$E_A = V_{\phi,OC} = \frac{V_T}{\sqrt{3}} \\ = \frac{540 \text{ V}}{\sqrt{3}} = 311.8 \text{ V}$$

The short-circuit current I_A is just equal to the line current, since the generator is Y-connected:

$$I_{A,SC} = I_{L,SC} = 300 \text{ A}$$

Therefore, the synchronous reactance at the rated field current can be calculated from Equation (5-25):

$$\sqrt{R_A^2 + X_S^2} = \frac{E_A}{I_A}$$

$$\sqrt{(0.2 \Omega)^2 + X_S^2} = \frac{311.8 \text{ V}}{300 \text{ A}}$$

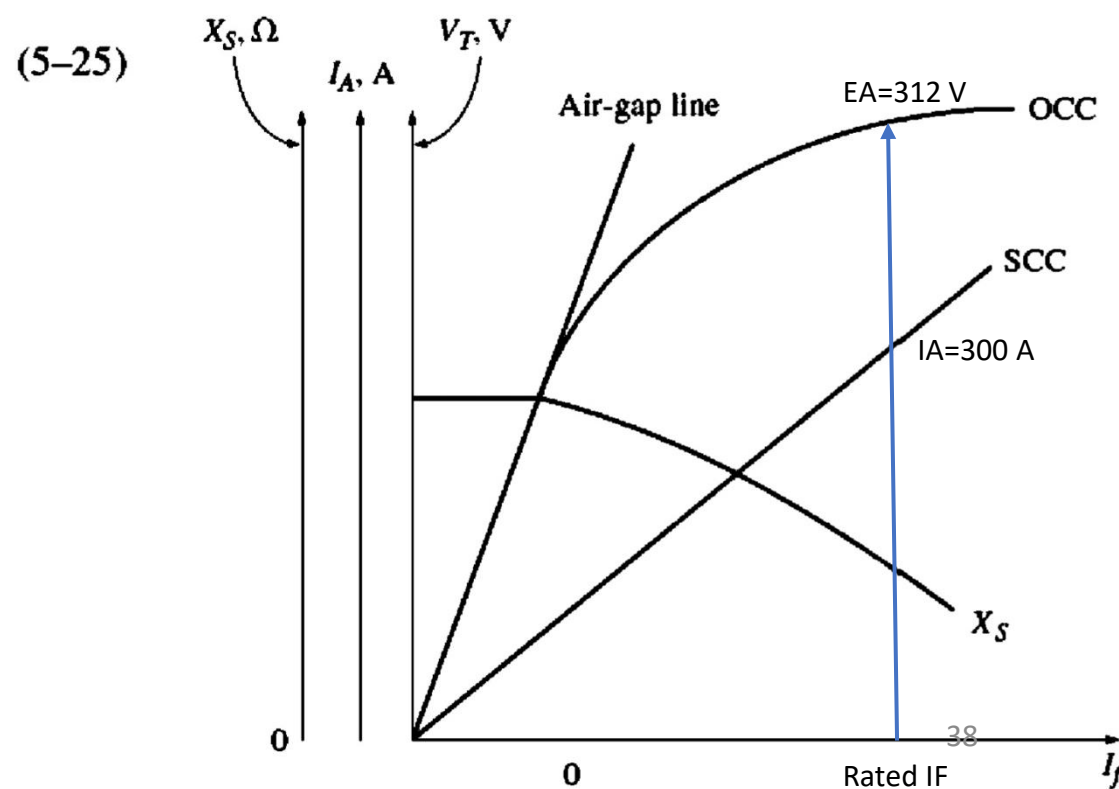
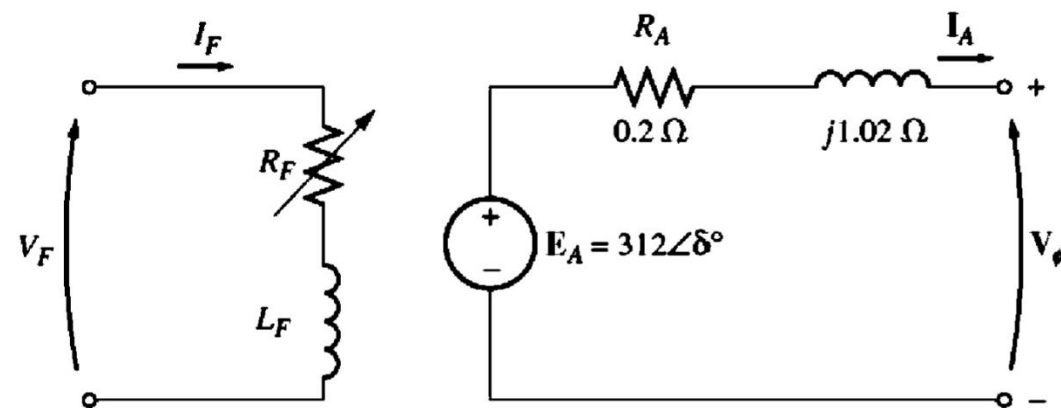
$$\sqrt{(0.2 \Omega)^2 + X_S^2} = 1.039 \Omega$$

$$0.04 + X_S^2 = 1.08$$

$$X_S^2 = 1.04$$

$$X_S = 1.02 \Omega$$

$$X_S = \frac{E_A}{I_A} = \frac{311.8 \text{ V}}{300 \text{ A}} = 1.04 \Omega \quad \text{Ignoring } R_A$$



Example 5–2. A 480-V, 60-Hz, Δ -connected, four-pole synchronous generator has the OCC shown in Figure 5–23a. This generator has a synchronous reactance of $0.1\ \Omega$ and an armature resistance of $0.015\ \Omega$. At full load, the machine supplies 1200 A at 0.8 PF lagging. Under full-load conditions, the friction and windage losses are 40 kW, and the core losses are 30 kW. Ignore any field circuit losses.

- (a) What is the speed of rotation of this generator?
- (b) How much field current must be supplied to the generator to make the terminal voltage 480 V at no load?
- (c) If the generator is now connected to a load and the load draws 1200 A at 0.8 PF lagging, how much field current will be required to keep the terminal voltage equal to 480 V?
- (d) How much power is the generator now supplying? How much power is supplied to the generator by the prime mover? What is this machine's overall efficiency?
- (e) If the generator's load were suddenly disconnected from the line, what would happen to its terminal voltage?
- (f) Finally, suppose that the generator is connected to a load drawing 1200 A at 0.8 PF *leading*. How much field current would be required to keep V_T at 480 V?

Solution:

This synchronous generator is Δ -connected, so its phase voltage is equal to its line voltage $V_\phi = V_T$, while its phase current is related to its line current by the equation $I_L = \sqrt{3}I_\phi$.

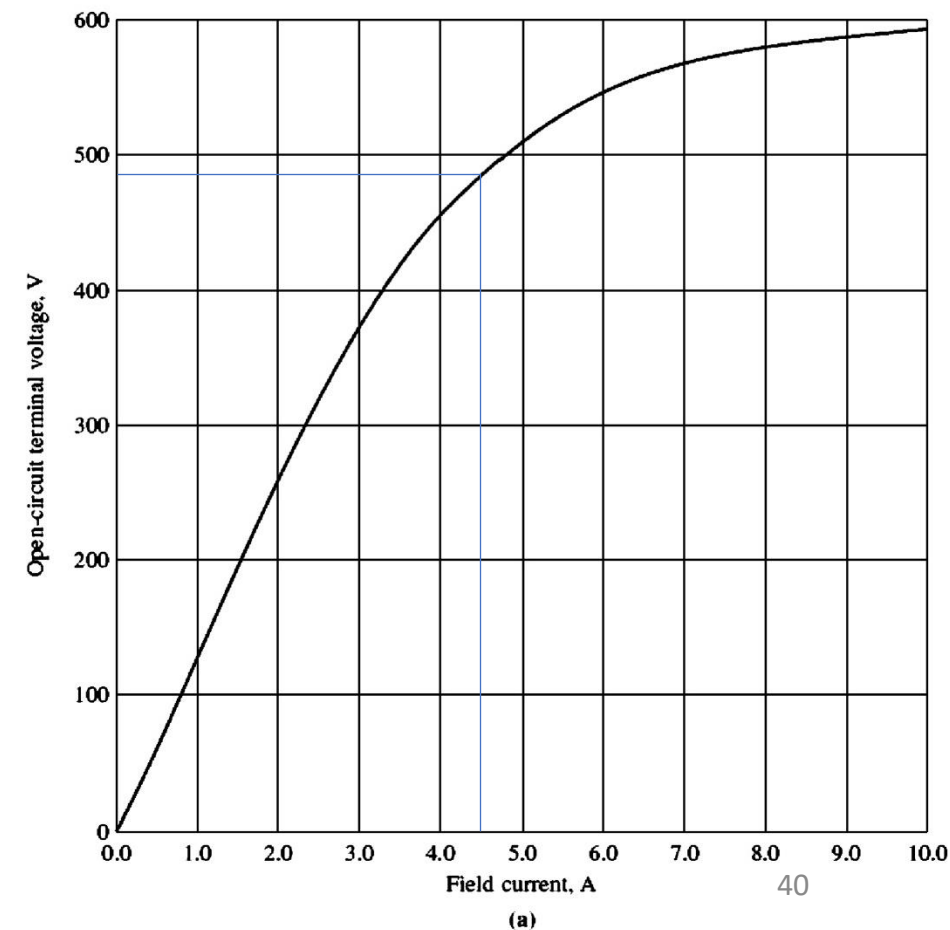
(a) The relationship between the electrical frequency produced by a synchronous generator and the mechanical rate of shaft rotation is given by Equation (4–34):

$$f_e = \frac{n_m P}{120} \quad (4-34)$$

Therefore,

$$\begin{aligned} n_m &= \frac{120f_e}{P} \\ &= \frac{120(60 \text{ Hz})}{4 \text{ poles}} = 1800 \text{ r/min} \end{aligned}$$

(b) In this machine, $V_T = V_\phi$. Since the generator is at no load, $I_A = 0$ and $E_A = V_\phi$. Therefore, $V_T = V_\phi = E_A = 480 \text{ V}$, and from the open-circuit characteristic, $I_F = 4.5 \text{ A}$.



(c) If the generator is supplying 1200 A, then the armature current in the machine is

$$I_A = \frac{1200 \text{ A}}{\sqrt{3}} = 692.8 \text{ A}$$

The phasor diagram for this generator is shown in Figure 5–23b. If the terminal voltage is adjusted to be 480 V, the size of the internal generated voltage E_A is given by

$$\begin{aligned} E_A &= V_\phi + R_A I_A + jX_S I_A \\ &= 480 \angle 0^\circ \text{ V} + (0.015 \Omega)(692.8 \angle -36.87^\circ \text{ A}) + (j0.1 \Omega)(692.8 \angle -36.87^\circ \text{ A}) \\ &= 480 \angle 0^\circ \text{ V} + 10.39 \angle -36.87^\circ \text{ V} + 69.28 \angle 53.13^\circ \text{ V} \\ &= 529.9 + j49.2 \text{ V} = 532 \angle 5.3^\circ \text{ V} \end{aligned}$$

To keep the terminal voltage at 480 V, E_A must be adjusted to 532 V. From Figure 5–23, the required field current is 5.7 A.

(d) The power that the generator is now supplying can be found from Equation (5–16):

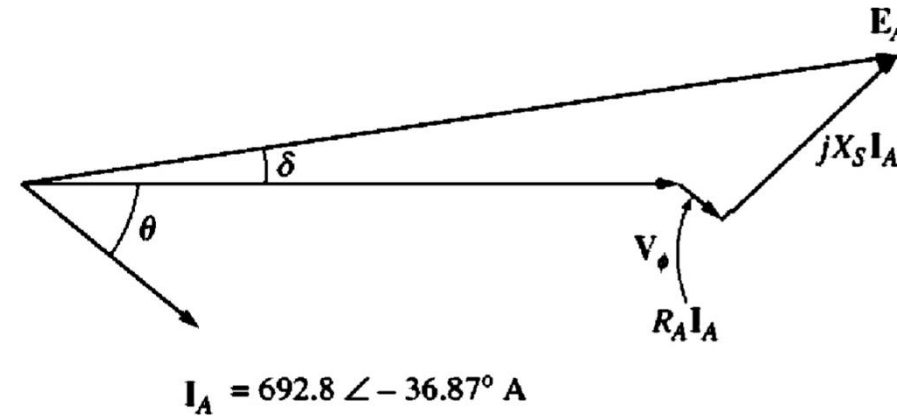
$$\begin{aligned} P_{\text{out}} &= \sqrt{3} V_T I_L \cos \theta \\ &= \sqrt{3}(480 \text{ V})(1200 \text{ A}) \cos 36.87^\circ \\ &= 798 \text{ kW} \end{aligned} \quad (5-16)$$

To determine the power input to the generator, use the power-flow diagram (Figure 5–15). From the power-flow diagram, the mechanical input power is given by

$$P_{\text{in}} = P_{\text{out}} + P_{\text{elec loss}} + P_{\text{core loss}} + P_{\text{mech loss}} + P_{\text{stray loss}}$$

The stray losses were not specified here, so they will be ignored. In this generator, the electrical losses are

$$\begin{aligned} P_{\text{elec loss}} &= 3I_A^2 R_A \\ &= 3(692.8 \text{ A})^2(0.015 \Omega) = 21.6 \text{ kW} \end{aligned}$$



The core losses are 30 kW, and the friction and windage losses are 40 kW, so the total input power to the generator is

$$P_{in} = 798 \text{ kW} + 21.6 \text{ kW} + 30 \text{ kW} + 40 \text{ kW} = 889.6 \text{ kW}$$

Therefore, the machine's overall efficiency is

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% = \frac{798 \text{ kW}}{889.6 \text{ kW}} \times 100\% = 89.75\%$$

(e) If the generator's load were suddenly disconnected from the line, the current I_A would drop to zero, making $E_A = V_\phi$. Since the field current has not changed, $|E_A|$ has not changed and V_ϕ and V_T must rise to equal E_A . Therefore, if the load were suddenly dropped, the terminal voltage of the generator would rise to 532 V.

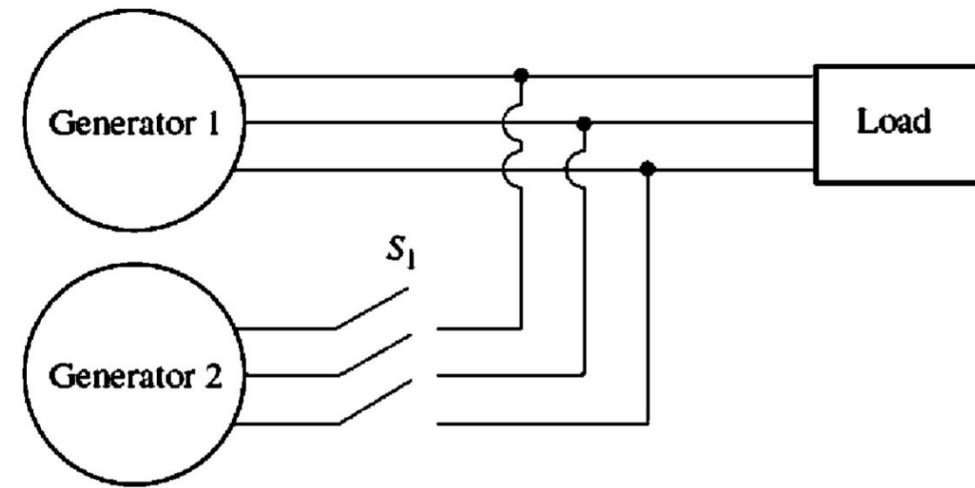
(f) If the generator were loaded down with 1200 A at 0.8 PF leading while the terminal voltage was 480 V, then the internal generated voltage would have to be

$$\begin{aligned} E_A &= V_\phi + R_A I_A + jX_S I_A \\ &= 480 \angle 0^\circ \text{ V} + (0.015 \Omega)(692.8 \angle 36.87^\circ \text{ A}) + (j0.1 \Omega)(692.8 \angle 36.87^\circ \text{ A}) \\ &= 480 \angle 0^\circ \text{ V} + 10.39 \angle 36.87^\circ \text{ V} + 69.28 \angle 126.87^\circ \text{ V} \\ &= 446.7 + j61.7 \text{ V} = 451 \angle 7.1^\circ \text{ V} \end{aligned}$$

Therefore, the internal generated voltage E_A must be adjusted to provide 451 V if V_T is to remain 480 V. Using the open-circuit characteristic, the field current would have to be adjusted to 4.1 A.

Example 5-3 –
self study

Parallel operation of SG (5.9)



1. Several generators can supply a bigger load than one machine by itself.
2. Having many generators increases the reliability of the power system, since the failure of any one of them does not cause a total power loss to the load.
3. Having many generators operating in parallel allows one or more of them to be removed for shutdown and preventive maintenance.
4. If only one generator is used and it is not operating at near full load, then it will be relatively inefficient. With several smaller machines in parallel, it is possible to operate only a fraction of them. The ones that do operate are operating near full load and thus more efficiently.

The conditions required for paralleling:

1. The rms *line voltages* of the two generators must be equal.
2. The two generators must have the same *phase sequence*.
3. The phase angles of the two *a* phases must be equal.
4. The frequency of the new generator, called the *oncoming generator*, must be slightly higher than the frequency of the running system.

Explanation:

These paralleling conditions require some explanation. Condition 1 is obvious—in order for two sets of voltages to be identical, they must of course have the same rms magnitude of voltage. The voltage in phases a and a' will be completely identical at all times if both their magnitudes and their angles are the same, which explains condition 3.

Condition 2 ensures that the sequence in which the phase voltages peak in the two generators is the same. If the phase sequence is different (as shown in Figure 5–27a), then even though one pair of voltages (the a phases) are in phase, the other two pairs of voltages are 120° out of phase. If the generators were connected in this manner, there would be no problem with phase a , but huge currents would flow in phases b and c , damaging both machines. To correct a phase sequence problem, simply swap the connections on any two of the three phases on one of the machines.

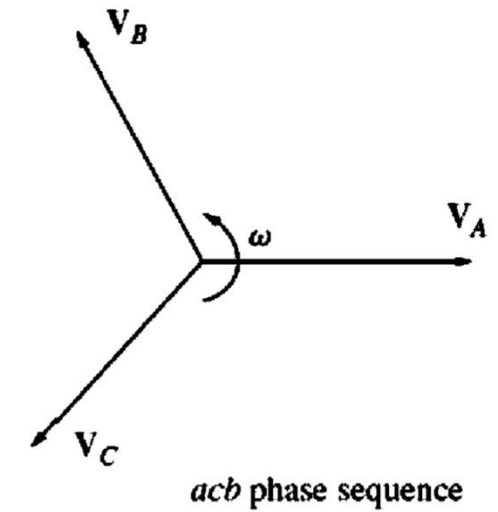
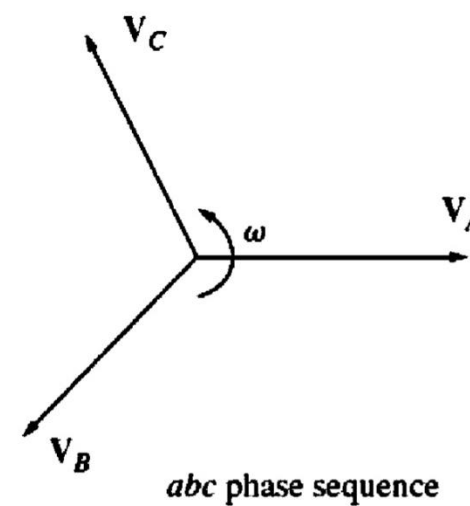
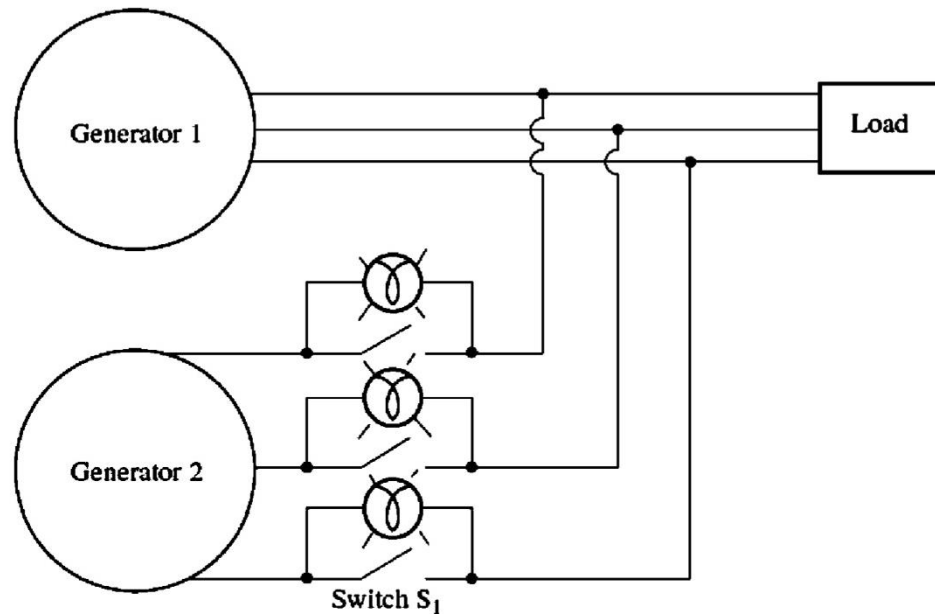
If the frequencies of the generators are not very nearly equal when they are connected together, large power transients will occur until the generators stabilize at a common frequency. The frequencies of the two machines must be very nearly equal, but they cannot be exactly equal. They must differ by a small amount so that the phase angles of the oncoming machine will change slowly with respect to the phase angles of the running system. In that way, the angles between the voltages can be observed and switch S_1 can be closed when the systems are exactly in phase.

Condition 1 and 3: Each phase voltage and phase angle must be identical.

Condition 2: Phase sequence must be identical. Failing to do so, there will be huge currents between phases.

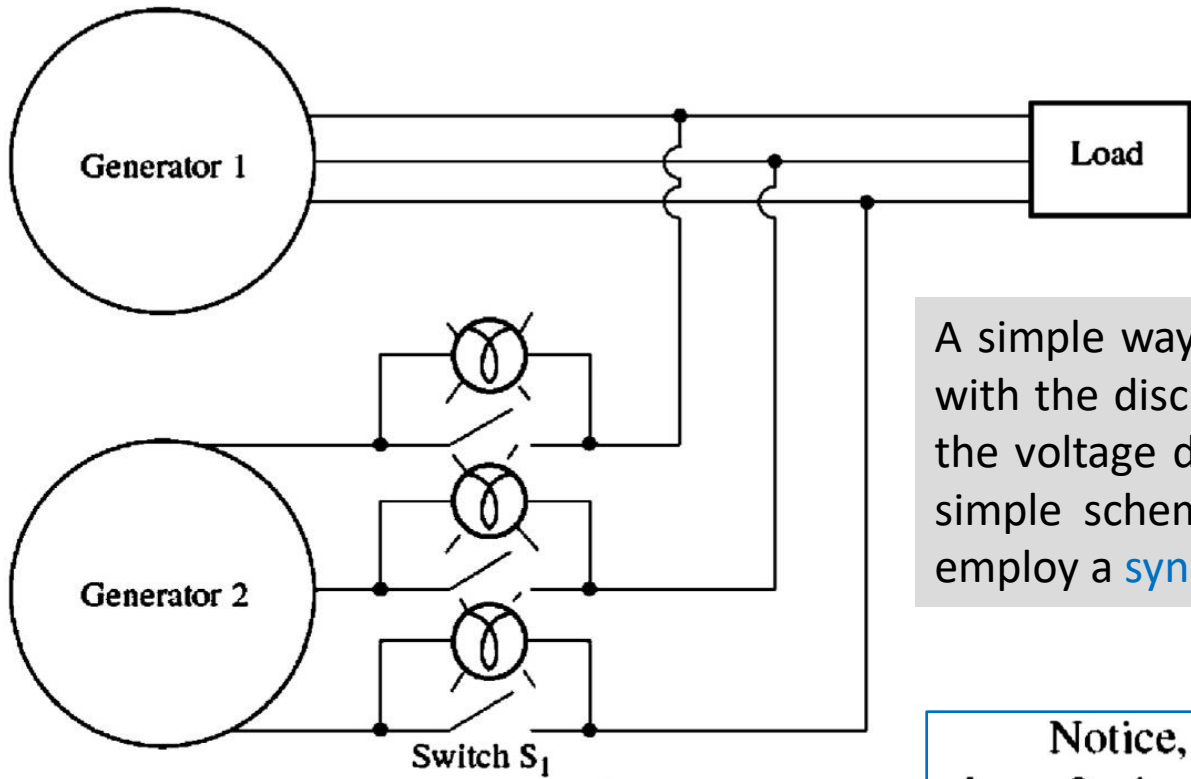
Condition 4: Frequency must be slightly higher to let the phase angle change according to the phase angle of the system. If the frequency is lower, large power transient will occur.

The general procedure for paralleling SG:



First	Using voltmeters, the field current of the oncoming generator should be adjusted until its terminal voltage is equal to the line voltage of the running system.
Second	The phase sequence of the oncoming generator must be compared to the phase sequence of the running system. The phase sequence can be checked in a number of different ways. One way is to alternately connect a small induction motor to the terminals of each of the two generators. Another way to check the phase sequence is the three-light-bulb method. In this approach, three light bulbs are stretched across the open terminals as shown in Figure. As the phase changes between the two systems, the light bulbs first get bright (large phase difference) and then get dim (small phase difference). If all three bulbs get bright and dark together, then the systems have the same phase sequence. If the bulbs brighten in succession, then the systems have the opposite phase sequence, and one of the sequences must be reversed.

Third	The frequency of the oncoming generator is adjusted to be slightly higher than the frequency of the running system. This is done first by watching a frequency meter until the frequencies are close and then by observing changes in phase between the systems. The oncoming generator is adjusted to a slightly higher frequency so that when it is connected, it will come on the line supplying power as a generator, instead of consuming it as a motor. Once the frequencies are very nearly equal, the voltages in the two systems will change phase with respect to each other very slowly, The phase changes are observed, and when the phase angles are equal, the switch connecting the two systems together is shut.
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How to be sure that the systems are in phase ?

A simple way is to watch the three light bulbs described above in connection with the discussion of phase sequence. When the three light bulbs all go out, the voltage difference across them is zero and the systems are in phase. This simple scheme works. but it is not very accurate. A better approach is to employ a [synchroscope](#).

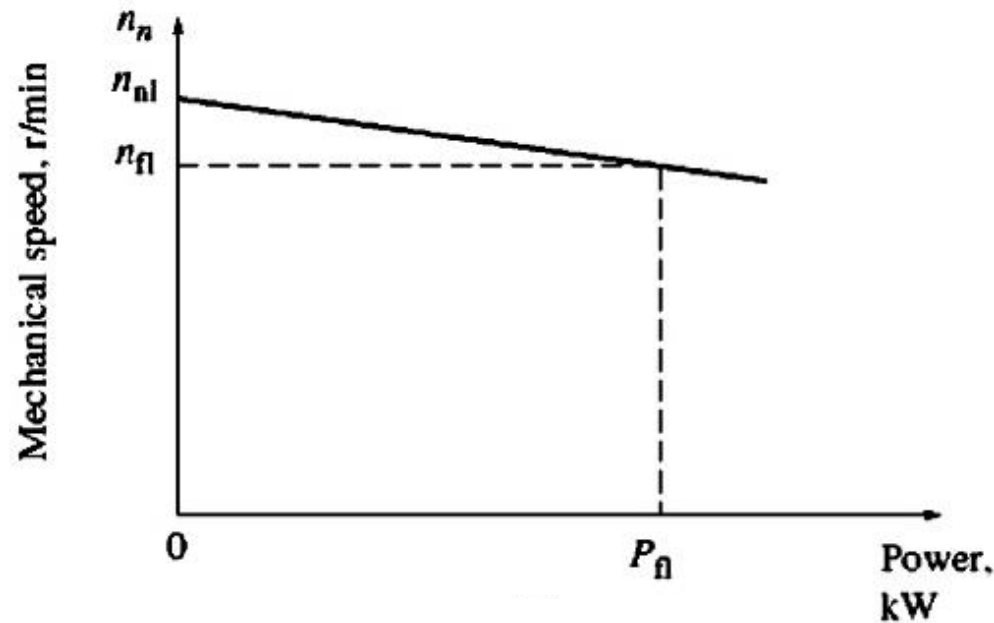
Notice, though, that a *synchroscope checks the relationships on only one phase*. It gives no information about phase sequence.

Frequency-power and voltage-reactive power characteristics of a SG:

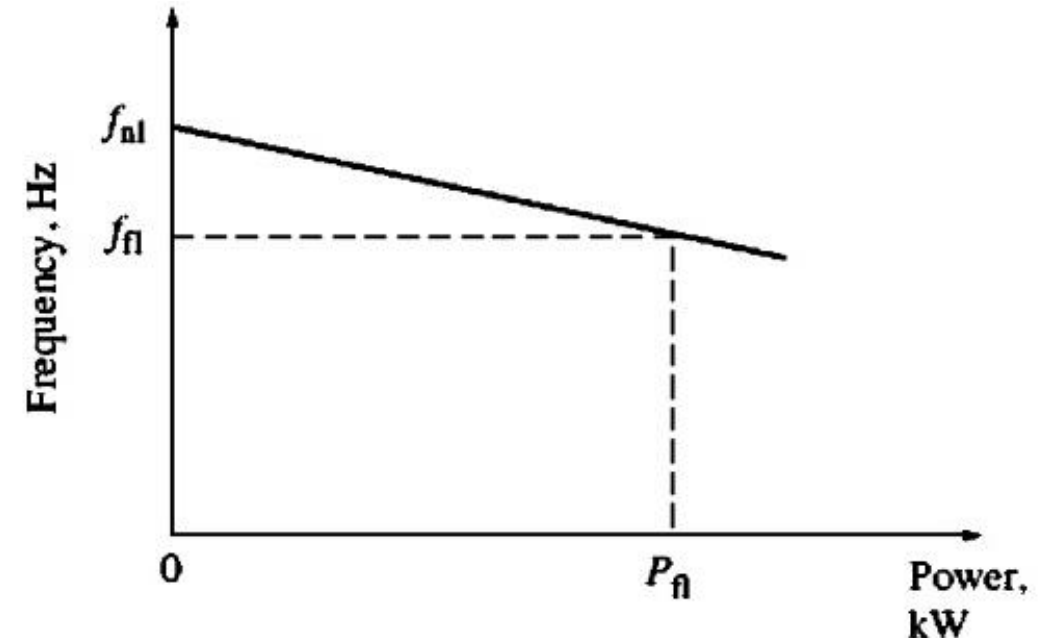
Whatever governor mechanism is present on a prime mover, it will always be adjusted to provide a slight drooping characteristic with increasing load. The speed droop (SD) of a prime mover is defined by the equation

$$SD = \frac{n_{nl} - n_{fl}}{n_{fl}} \times 100\%$$

where n_{nl} is the no-load prime-mover speed and n_{fl} is the full-load prime-mover speed. Most generator prime movers have a speed droop of 2 to 4 percent, as defined in Equation (5-27).



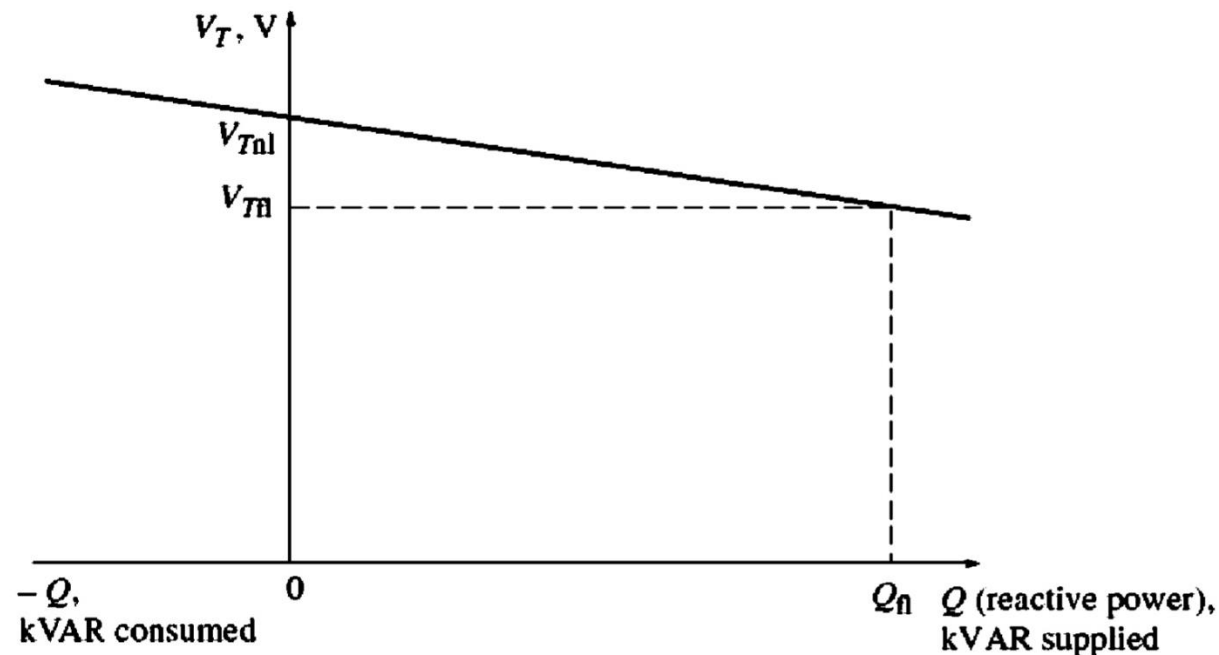
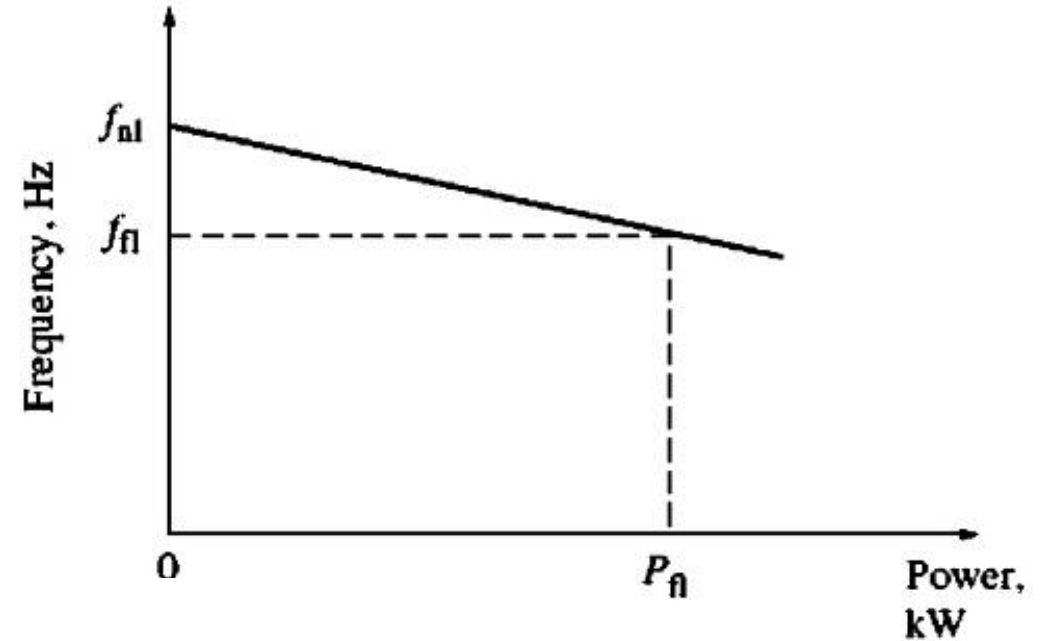
$$f_e = \frac{n_m P}{120}$$



The relationship between frequency and power can be described quantitatively by the equation

$$P = s_P(f_{nl} - f_{sys})$$

where P = power output of the generator
 f_{nl} = no-load frequency of the generator
 f_{sys} = operating frequency of system
 s_P = slope of curve, in kW/Hz or MW/Hz



This characteristics is not intrinsically linear, but many generator voltage regulators include a feature to make it so.

To summarize, when a generator is operating by itself supplying the system loads, then

1. The real and reactive power supplied by the generator will be the amount demanded by the attached load.
2. The governor set points of the generator will control the operating frequency of the power system.
3. The field current (or the field regulator set points) control the terminal voltage of the power system.

This is the situation found in isolated generators in remote field environments.

Operation of SG in parallel with large POWER SYSTEM

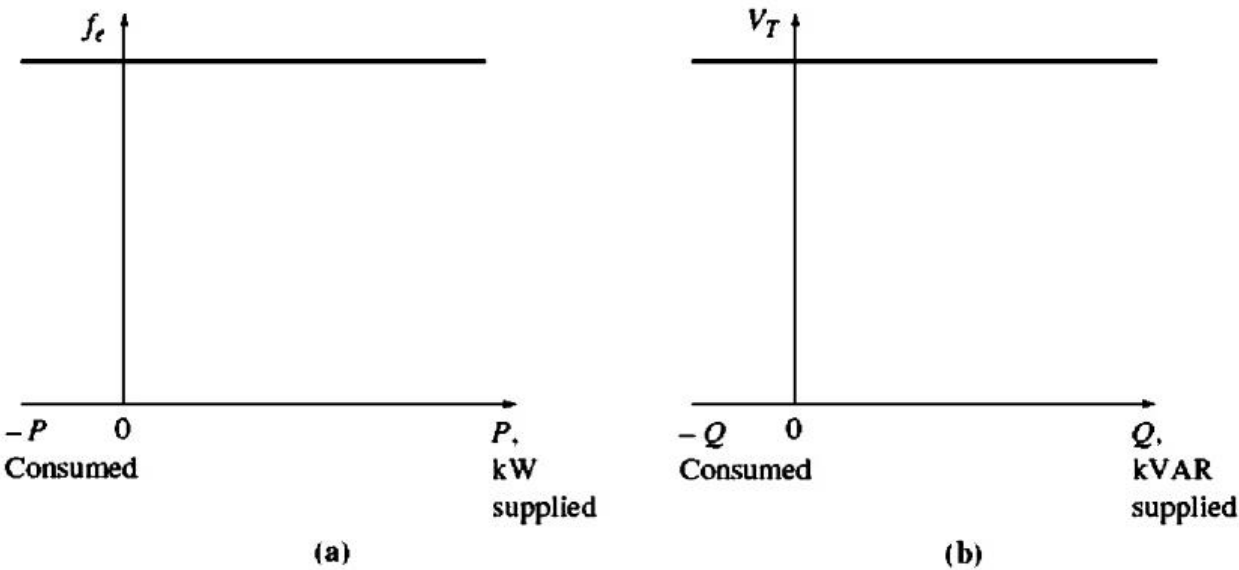
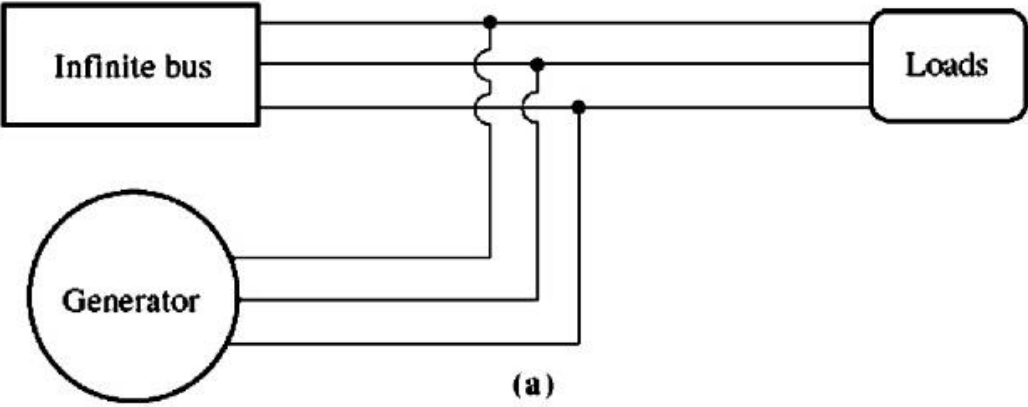
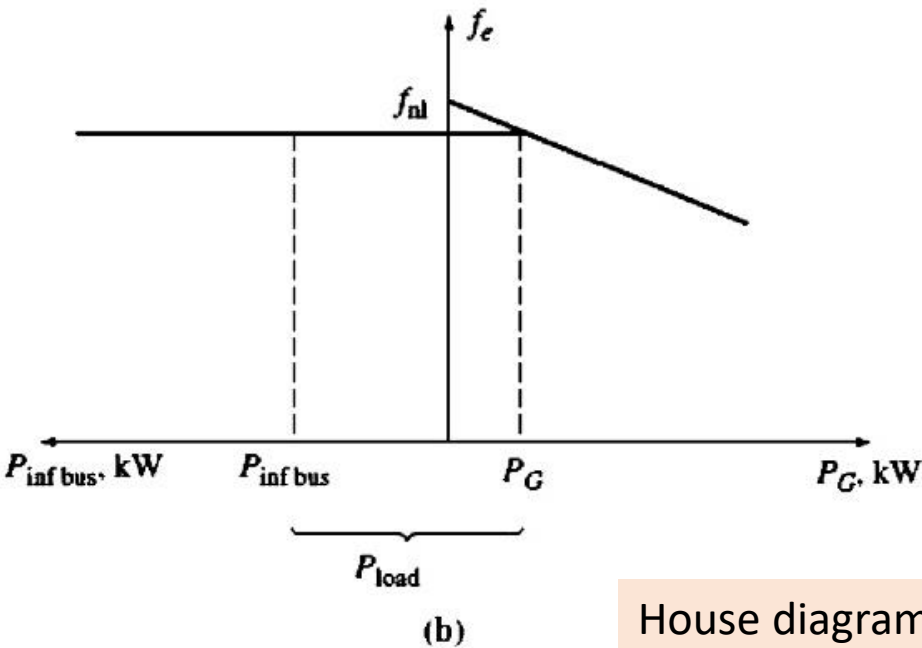


FIGURE 5–32
Curves for an infinite bus: (a) frequency versus power and (b) terminal voltage versus reactive power.



House diagram

FIGURE 5–33
(a) A synchronous generator operating in parallel with an infinite bus. (b) The frequency-versus-power diagram (or *house diagram*) for a synchronous generator in parallel with an infinite bus.

Assume that the generator has just been paralleled with the infinite bus according to the procedure described previously. Then the generator will be essentially “floating” on the line, supplying a small amount of real power and little or no reactive power. This situation is shown in Figure 5–34.

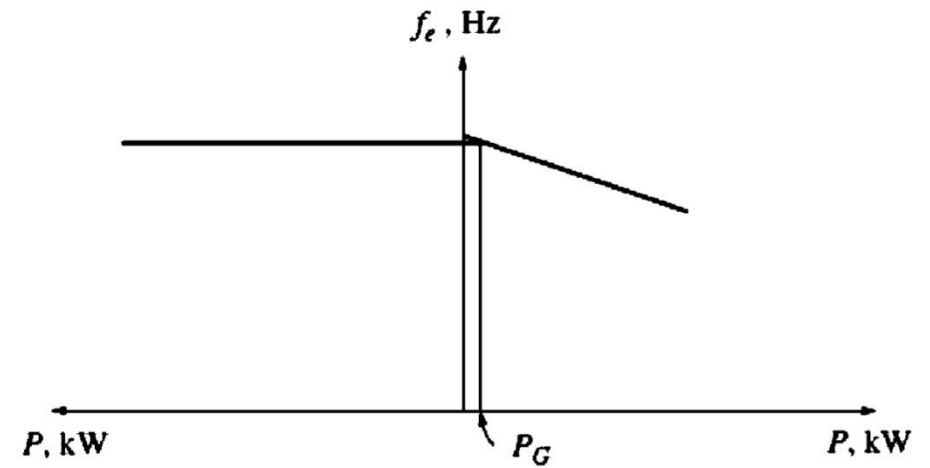


FIGURE 5–34
The frequency-versus-power diagram at the moment just after paralleling.

Assume that the generator has just been paralleled with the infinite bus according to the procedure described previously. Then the generator will be essentially “floating” on the line, supplying a small amount of real power and little or no reactive power. This situation is shown in Figure 5–34.

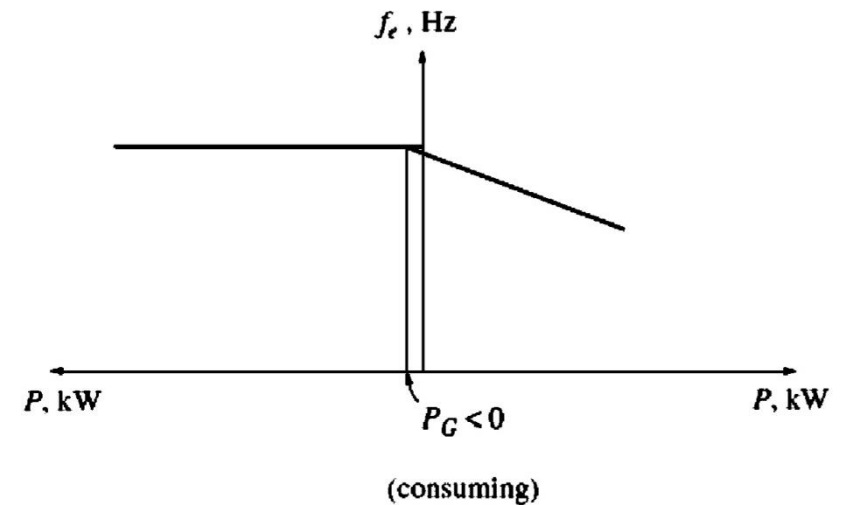
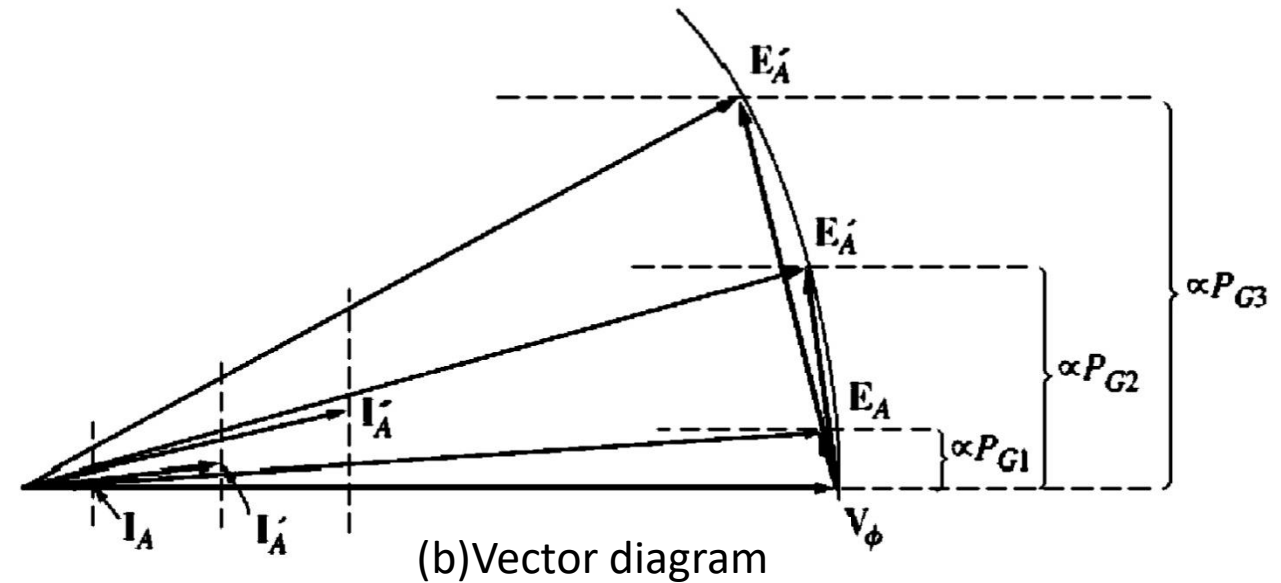
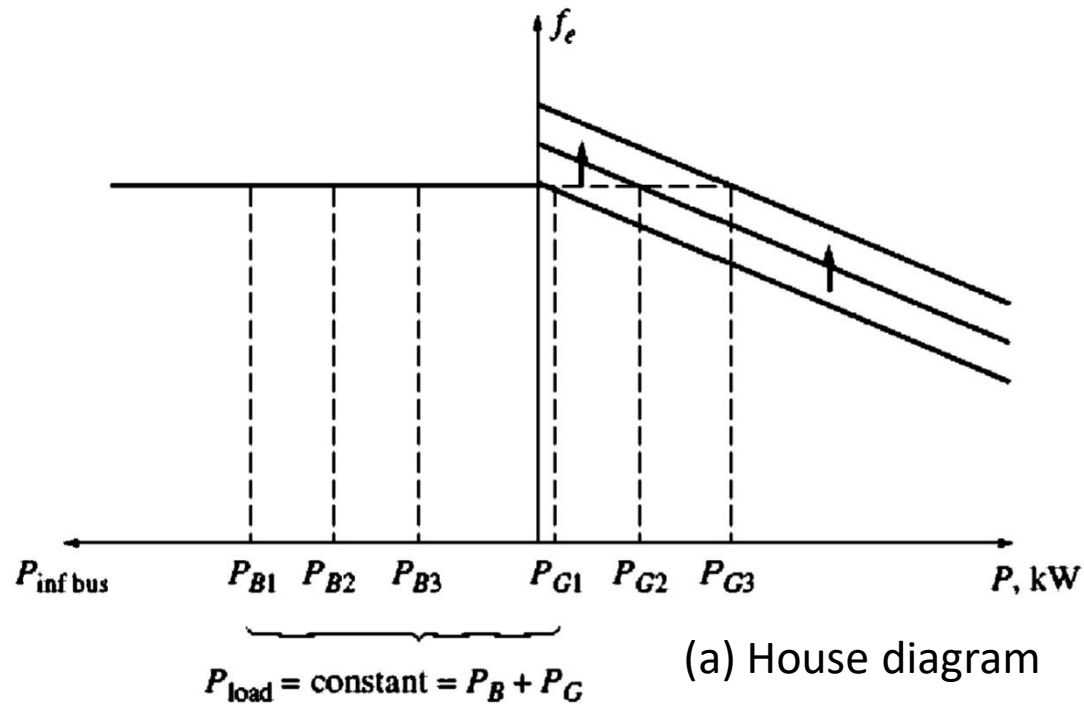


FIGURE 5–35
The frequency-versus-power diagram if the no-load frequency of the generator were slightly *less* than system frequency before paralleling.

Once the generator has been connected, what happens when its governor set points are increased?



This is shown by the house diagram in Figure 5–36a and by the phasor diagram in Figure 5–36b. Notice in the phasor diagram that $E_A \sin \delta$ (which is proportional to the power supplied as long as V_T is constant) has increased, while the magnitude of $E_A (= K\phi\omega)$ remains constant, since both the field current I_F and the speed of rotation ω are unchanged.

If the power supplied is constant as the field current is changed, then the distances proportional to the power in the phasor diagram ($I_A \cos \theta$ and $E_A \sin \delta$) cannot change. When the field current is increased, the flux ϕ increases, and therefore $E_A (= K\phi\omega)$ increases. If E_A increases, but $E_A \sin \delta$ must remain constant, then the phasor E_A must “slide” along the line of constant power, as shown in Figure 5–37. Since V_ϕ is constant, the angle of $jX_S I_A$ changes as shown, and therefore the angle and magnitude of I_A change. Notice that as a result the distance proportional to Q ($I_A \sin \theta$) increases. In other words, *increasing the field current in a synchronous generator operating in parallel with an infinite bus increases the reactive power output of the generator.*

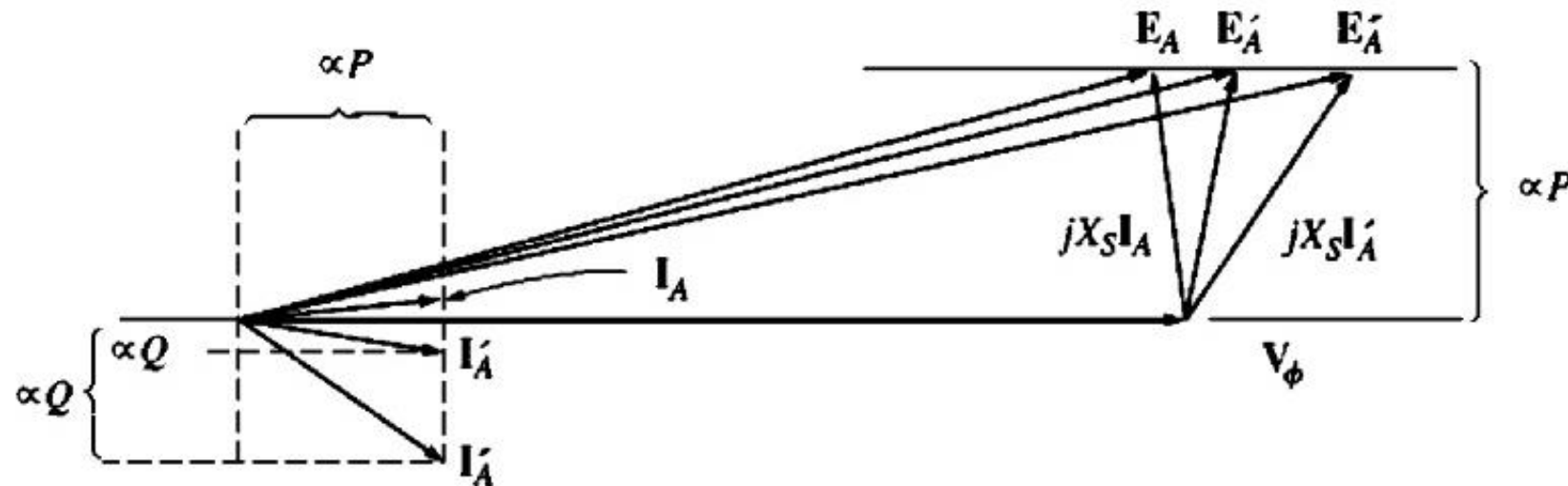


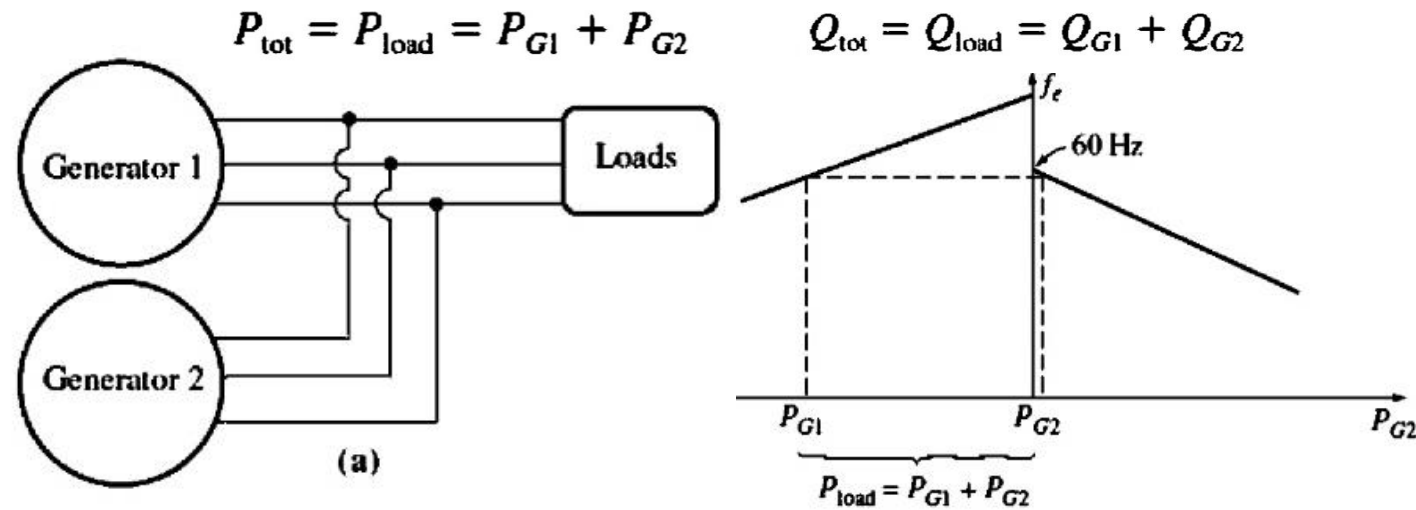
FIGURE 5–37

The effect of increasing the generator's field current on the phasor diagram of the machine.

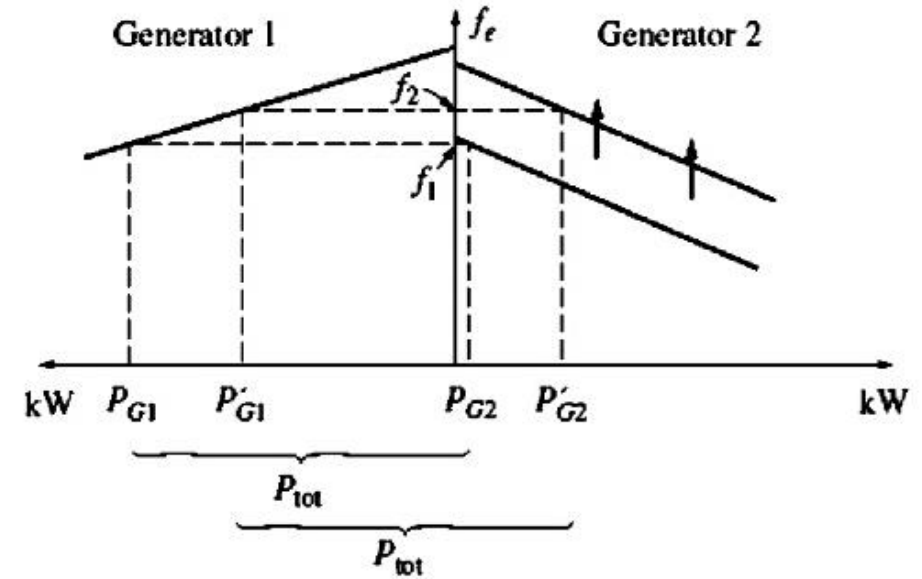
To summarize, when a generator is operating in parallel with an infinite bus:

- 1. The frequency and terminal voltage of the generator are controlled by the system to which it is connected.**
- 2. The governor set points of the generator control the real power supplied by the generator to the system.**
- 3. The field current in the generator controls the reactive power supplied by the generator to the system.**

Operation of SG in parallel with other generators of the same size



Ref [3]: Electric Machinery Fundamentals (4th Edition) - Stephen J. Chapman

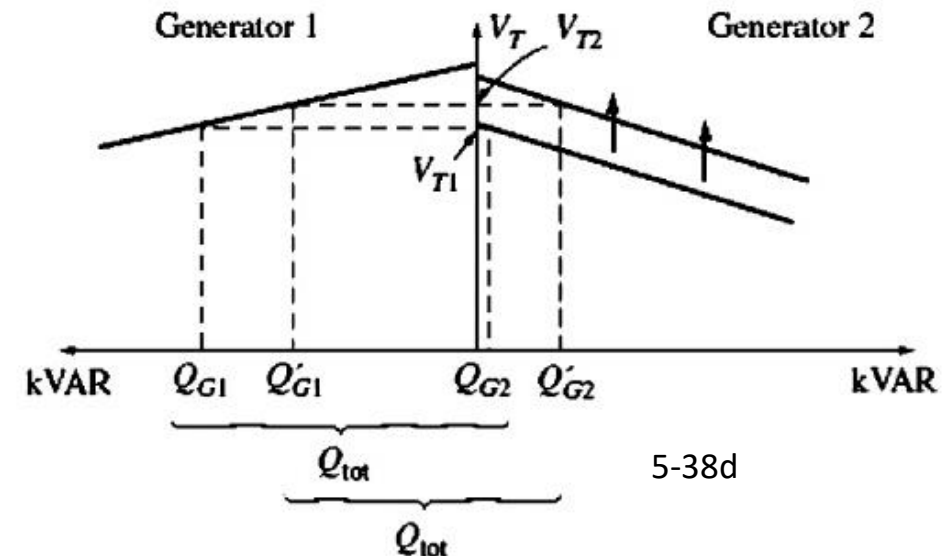


Therefore, when two generators are operating together, an increase in governor set points on one of them

1. *Increases the system frequency.*
2. *Increases the power supplied by that generator, while reducing the power supplied by the other one.*

What happens if the field current of G_2 is increased? The resulting behavior is analogous to the real-power situation and is shown in Figure 5–38d. When two generators are operating together and the field current of G_2 is increased,

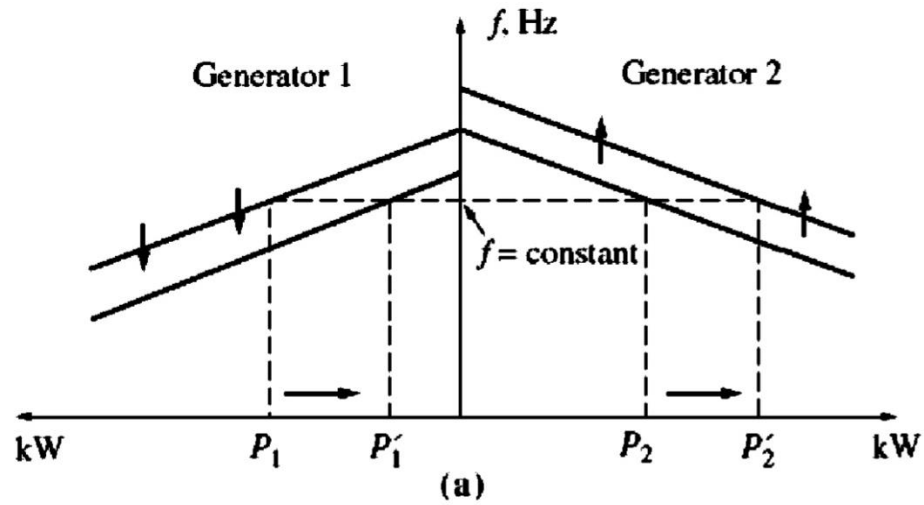
1. *The system terminal voltage is increased.*
2. *The reactive power Q supplied by that generator is increased, while the reactive power supplied by the other generator is decreased.*



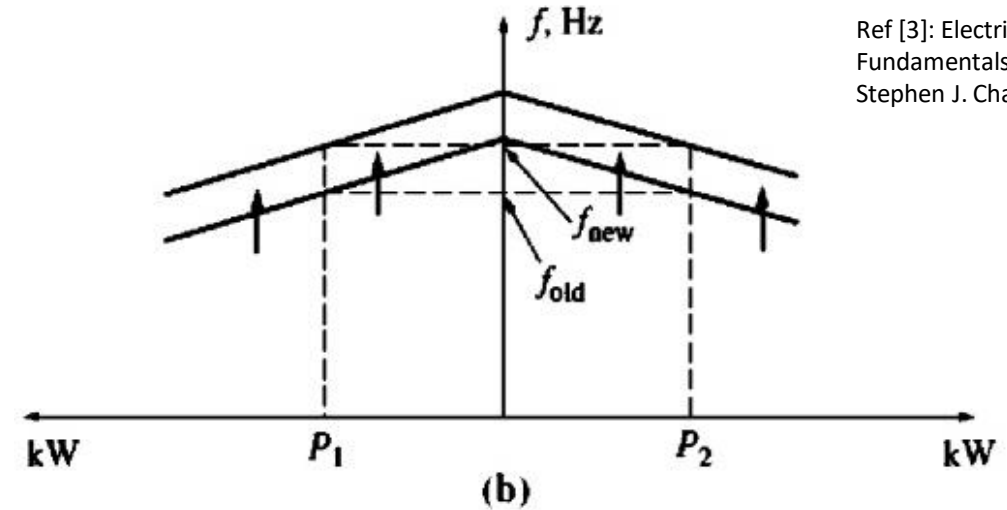
5-38d

How can the power sharing of the power system be adjusted independently of the system frequency?

Ref [3]: Electric Machinery Fundamentals (4th Edition) - Stephen J. Chapman

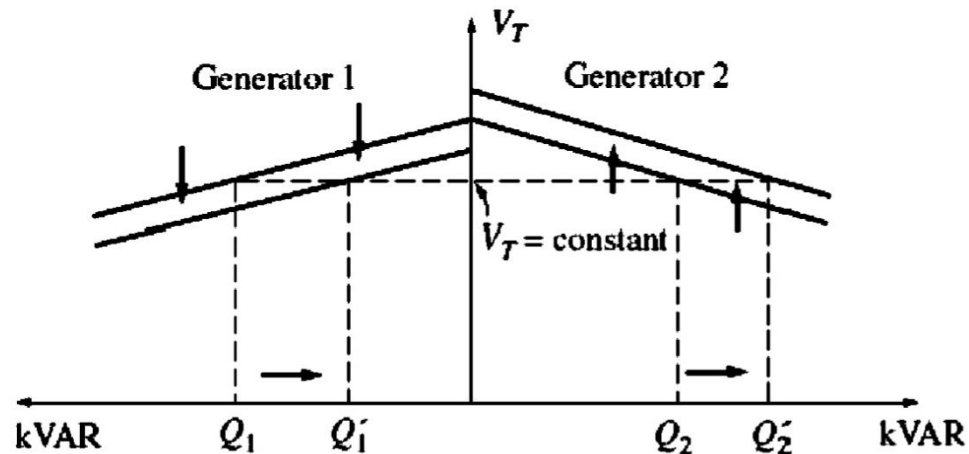


Frequency fixed, P_1 and P_2 changed

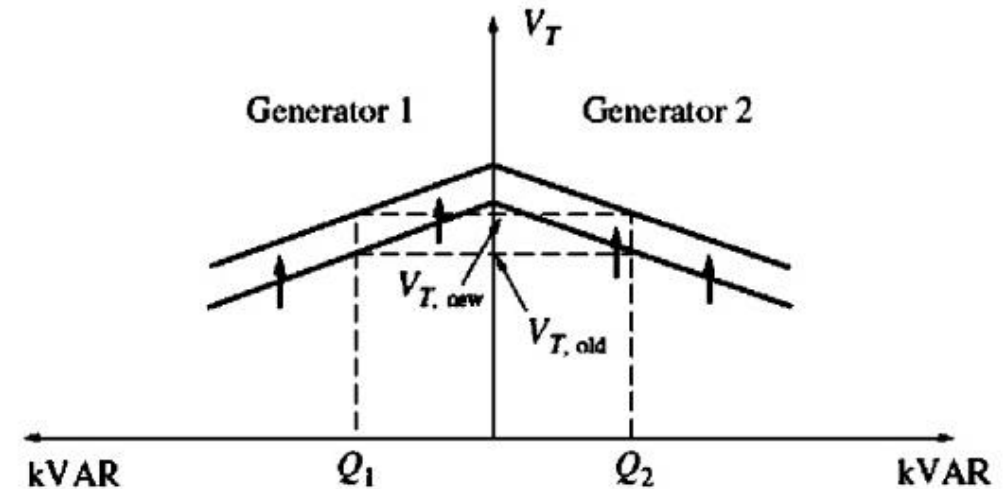


Frequency fixed, P_1 and P_2 fixed

How can the reactive power sharing of the power system be adjusted independently of the terminal voltage?



V_T fixed, Q_1 and Q_2 changed



V_T changed, Q_1 and Q_2 fixed

To summarize, in the case of two generators operating together:

1. The system is constrained in that the total power supplied by the two generators together must equal the amount consumed by the load. Neither f_{sys} nor V_T is constrained to be constant.
2. To adjust the real power sharing between generators without changing f_{sys} , simultaneously increase the governor set points on one generator while decreasing the governor set points on the other. The machine whose governor set point was increased will assume more of the load.
3. To adjust f_{sys} without changing the real power sharing, simultaneously increase or decrease both generators' governor set points.
4. To adjust the reactive power sharing between generators without changing V_T , simultaneously increase the field current on one generator while decreasing the field current on the other. The machine whose field current was increased will assume more of the reactive load.
5. To adjust V_T without changing the reactive power sharing, simultaneously increase or decrease both generators' field currents.

Overall summary of sg operation

SG Operation	Control parameters	Change in P	Change in Q	Change in f	Change in Vt
Alone	Governor	No. P is demanded by load	No. Q is demanded by load	Yes	No
	Excitation			No	Yes
Connected to infinite bus bar	Governor	Yes	No	No. f of infinite bus is fixed.	No
	Excitation	No	Yes	No	No. Vt of Infinite bus is fixed
Connected to another SG of same size	Governor	Yes	No	Yes	No
	Excitation	No	Yes	No	Yes

Math Practice sg2

Example 5–5. Figure 5–31 shows a generator supplying a load. A second load is to be connected in parallel with the first one. The generator has a no-load frequency of 61.0 Hz and a slope s_p of 1 MW/Hz. Load 1 consumes a real power of 1000 kW at 0.8 PF lagging, while load 2 consumes a real power of 800 kW at 0.707 PF lagging.

- (a) Before the switch is closed, what is the operating frequency of the system?
- (b) After load 2 is connected, what is the operating frequency of the system?
- (c) After load 2 is connected, what action could an operator take to restore the system frequency to 60 Hz?

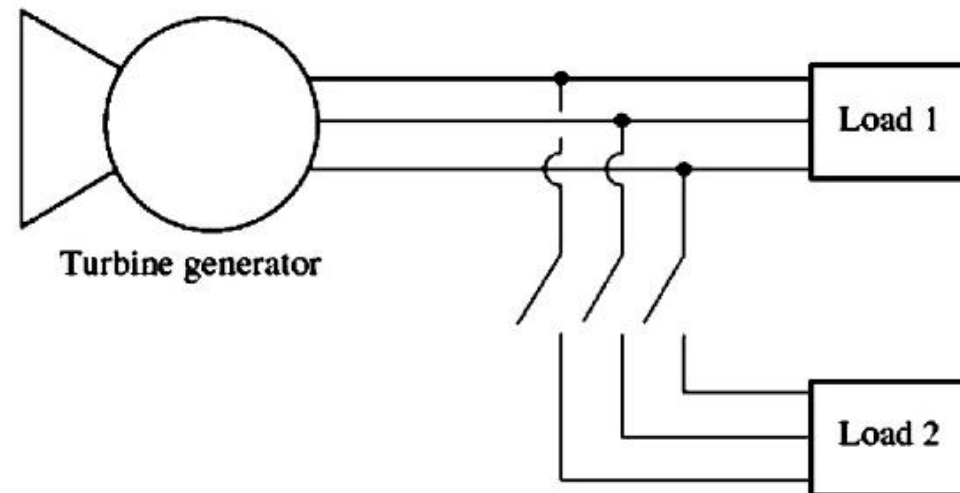


FIGURE 5–31

The power system in Example 5–5.

Solution

This problem states that the slope of the generator's characteristic is 1 MW/Hz and that its no-load frequency is 61 Hz. Therefore, the power produced by the generator is given by

$$P = s_P(f_{nl} - f_{sys}) \quad (5-28)$$

so
$$f_{sys} = f_{nl} - \frac{P}{s_P}$$

(a) The initial system frequency is given by

$$\begin{aligned} f_{sys} &= f_{nl} - \frac{P}{s_P} \\ &= 61 \text{ Hz} - \frac{1000 \text{ kW}}{1 \text{ MW/Hz}} = 61 \text{ Hz} - 1 \text{ Hz} = 60 \text{ Hz} \end{aligned}$$

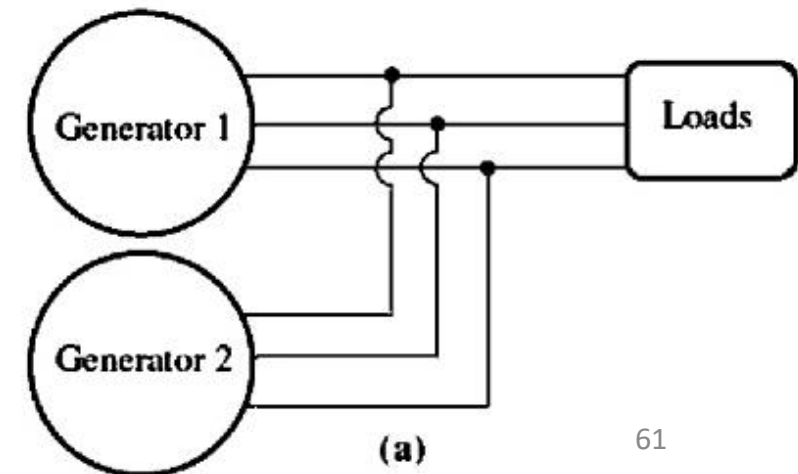
(b) After load 2 is connected,

$$\begin{aligned} f_{sys} &= f_{nl} - \frac{P}{s_P} \\ &= 61 \text{ Hz} - \frac{1800 \text{ kW}}{1 \text{ MW/Hz}} = 61 \text{ Hz} - 1.8 \text{ Hz} = 59.2 \text{ Hz} \end{aligned}$$

(c) After the load is connected, the system frequency falls to 59.2 Hz. To restore the system to its proper operating frequency, the operator should increase the governor no-load set points by 0.8 Hz, to 61.8 Hz. This action will restore the system frequency to 60 Hz.

Example 5–6. Figure 5–38a shows two generators supplying a load. Generator 1 has a no-load frequency of 61.5 Hz and a slope s_{p1} of 1 MW/Hz. Generator 2 has a no-load frequency of 61.0 Hz and a slope s_{p2} of 1 MW/Hz. The two generators are supplying a real load totaling 2.5 MW at 0.8 PF lagging. The resulting system power-frequency or house diagram is shown in Figure 5–39.

- (a) At what frequency is this system operating, and how much power is supplied by each of the two generators?
- (b) Suppose an additional 1-MW load were attached to this power system. What would the new system frequency be, and how much power would G_1 and G_2 supply now?
- (c) With the system in the configuration described in part *b*, what will the system frequency and generator powers be if the governor set points on G_2 are increased by 0.5 Hz?



Solution:

- (a) In the first case, both generators have a slope of 1 MW/Hz, and G_1 has a no-load frequency of 61.5 Hz, while G_2 has a no-load frequency of 61.0 Hz. The total load is 2.5 MW. Therefore, the system frequency can be found as follows:

$$P_1 = s_{P1}(f_{nl,1} - f_{sys})$$

$$P_2 = s_{P2}(f_{nl,2} - f_{sys})$$

$$P_{load} = P_1 + P_2$$

$$P_{load} = P_1 + P_2$$

$$= s_{P1}(f_{nl,1} - f_{sys}) + s_{P2}(f_{nl,2} - f_{sys})$$

$$\begin{aligned} 2.5 \text{ MW} &= (1 \text{ MW/Hz})(61.5 \text{ Hz} - f_{sys}) + (1 \text{ MW/Hz})(61 \text{ Hz} - f_{sys}) \\ &= 61.5 \text{ MW} - (1 \text{ MW/Hz})f_{sys} + 61 \text{ MW} - (1 \text{ MW/Hz})f_{sys} \\ &= 122.5 \text{ MW} - (2 \text{ MW/Hz})f_{sys} \end{aligned}$$

$$\text{therefore } f_{sys} = \frac{122.5 \text{ MW} - 2.5 \text{ MW}}{(2 \text{ MW/Hz})} = 60.0 \text{ Hz}$$

The resulting powers supplied by the two generators are

$$P_1 = s_{P1}(f_{nl,1} - f_{sys})$$

$$= (1 \text{ MW/Hz})(61.5 \text{ Hz} - 60.0 \text{ Hz}) = 1.5 \text{ MW}$$

$$P_2 = s_{P2}(f_{nl,2} - f_{sys})$$

$$= (1 \text{ MW/Hz})(61.0 \text{ Hz} - 60.0 \text{ Hz}) = 1 \text{ MW}$$

(b) When the load is increased by 1 MW, the total load becomes 3.5 MW. The new system frequency is now given by

$$P_{\text{load}} = s_{P1}(f_{\text{nl},1} - f_{\text{sys}}) + s_{P2}(f_{\text{nl},2} - f_{\text{sys}})$$

$$\begin{aligned} 3.5 \text{ MW} &= (1 \text{ MW/Hz})(61.5 \text{ Hz} - f_{\text{sys}}) + (1 \text{ MW/Hz})(61 \text{ Hz} - f_{\text{sys}}) \\ &= 61.5 \text{ MW} - (1 \text{ MW/Hz})f_{\text{sys}} + 61 \text{ MW} - (1 \text{ MW/Hz})f_{\text{sys}} \\ &= 122.5 \text{ MW} - (2 \text{ MW/Hz})f_{\text{sys}} \end{aligned}$$

$$\text{therefore } f_{\text{sys}} = \frac{122.5 \text{ MW} - 3.5 \text{ MW}}{(2 \text{ MW/Hz})} = 59.5 \text{ Hz}$$

The resulting powers are

$$\begin{aligned} P_1 &= s_{P1}(f_{\text{nl},1} - f_{\text{sys}}) \\ &= (1 \text{ MW/Hz})(61.5 \text{ Hz} - 59.5 \text{ Hz}) = 2.0 \text{ MW} \end{aligned}$$

$$\begin{aligned} P_2 &= s_{P2}(f_{\text{nl},2} - f_{\text{sys}}) \\ &= (1 \text{ MW/Hz})(61.0 \text{ Hz} - 59.5 \text{ Hz}) = 1.5 \text{ MW} \end{aligned}$$

$$P_1 = s_{P1}(f_{\text{nl},1} - f_{\text{sys}})$$

$$P_2 = s_{P2}(f_{\text{nl},2} - f_{\text{sys}})$$

$$P_{\text{load}} = P_1 + P_2$$

$$P_1 = s_{P1}(f_{nl,1} - f_{sys})$$

$$P_2 = s_{P2}(f_{nl,2} - f_{sys})$$

$$P_{load} = P_1 + P_2$$

(c) If the no-load governor set points of G_2 are increased by 0.5 Hz, the new system frequency becomes

$$P_{load} = s_{P1}(f_{nl,1} - f_{sys}) + s_{P2}(f_{nl,2} - f_{sys})$$

$$\begin{aligned} 3.5 \text{ MW} &= (1 \text{ MW/Hz})(61.5 \text{ Hz} - f_{sys}) + (1 \text{ MW/Hz})(61.5 \text{ Hz} - f_{sys}) \\ &= 123 \text{ MW} - (2 \text{ MW/Hz})f_{sys} \end{aligned}$$

$$f_{sys} = \frac{123 \text{ MW} - 3.5 \text{ MW}}{(2 \text{ MW/Hz})} = 59.75 \text{ Hz}$$

The resulting powers are

$$\begin{aligned} P_1 &= P_2 = s_{P1}(f_{nl,1} - f_{sys}) \\ &= (1 \text{ MW/Hz})(61.5 \text{ Hz} - 59.75 \text{ Hz}) = 1.75 \text{ MW} \end{aligned}$$

Hands on Winding design (FYI)

Draw the developed winding diagram of a 3 phase induction motor with 18 slots, 2 poles, single layer, full pitched winding.

Soln: No. of slots per pole per phase = $18/(2 \times 3) = 3$

Pole pitch = no. of conductor / pole = $18/2 = 9$

Slot angle = $180/\text{pole pitch} = 180/9 = 20$

Full pitched winding = coil span = 180

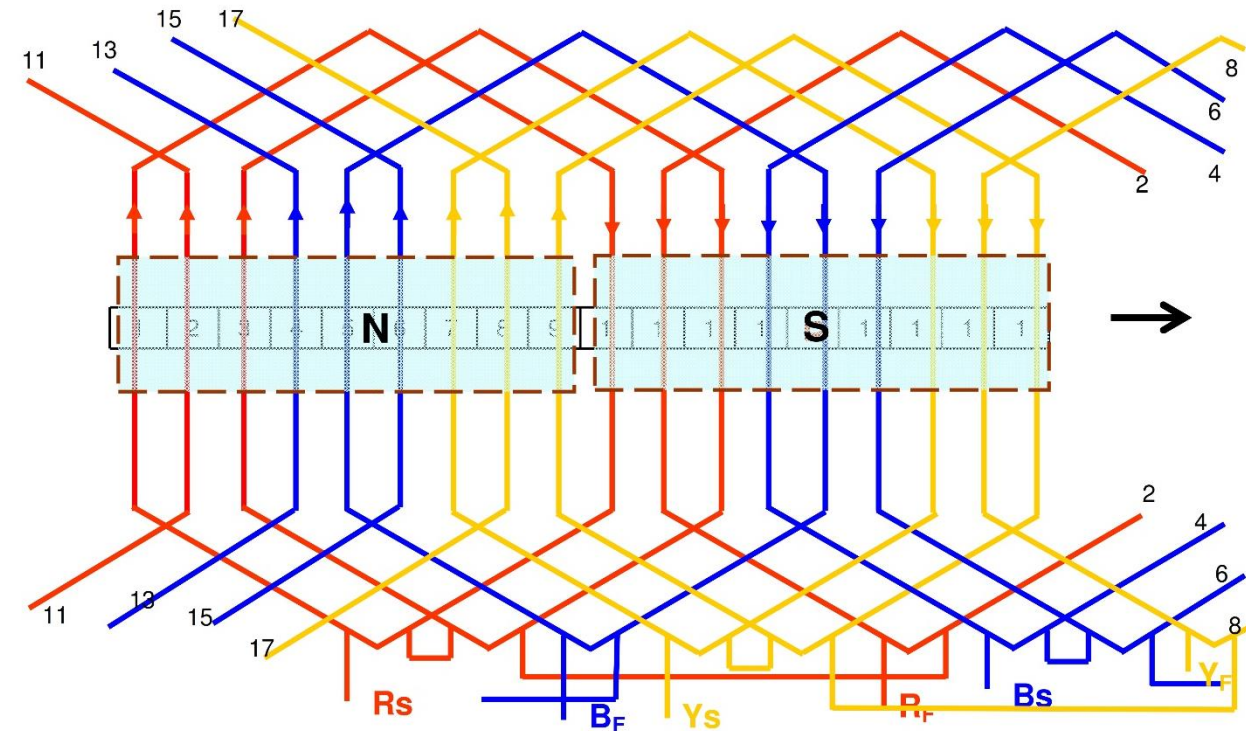
Coil span = winding pitch/slot angle = $180/20 = 9$ slots

Winding Table:

Phase	1st pole	2nd pole
R	$1 + 9 = 10$ $3 + 9 = 12$	$11 + 9 = 20 (2)$
B	$5 + 9 = 14$	$13 + 9 = 22(4)$ $15 + 9 = 24 (6)$
Y	$7 + 9 = 16$ $9 + 9 = 18$	$17 + 9 = 26 (8)$

Connections: $R_s = 1$, $Y_s = 1 + 120/\text{slot angle} = 1 + 120/20 = 7$; $B_s = 1 + 240/\text{slot angle} = 1 + 240/20 = 13$

Winding Diagram



Winding Diagram

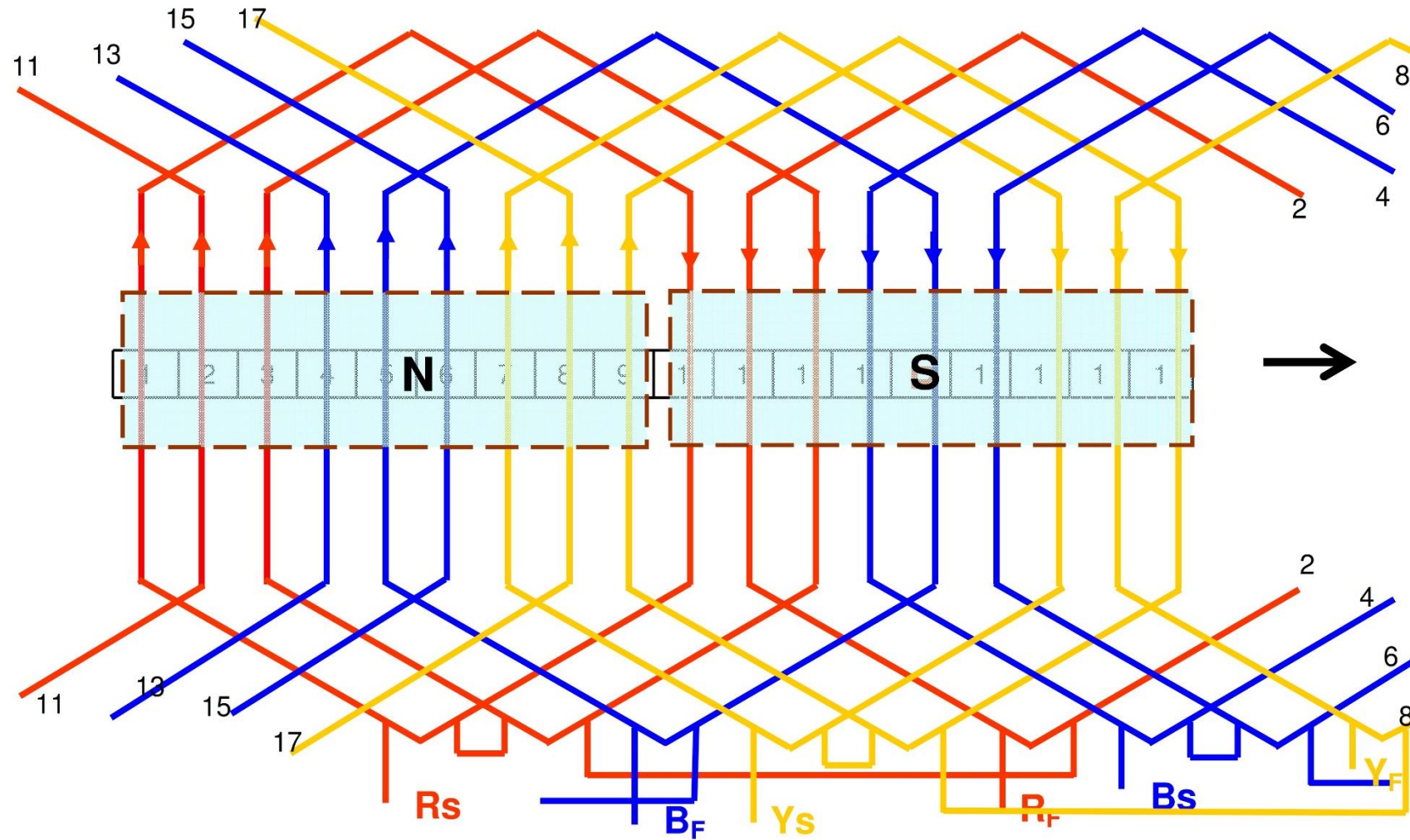
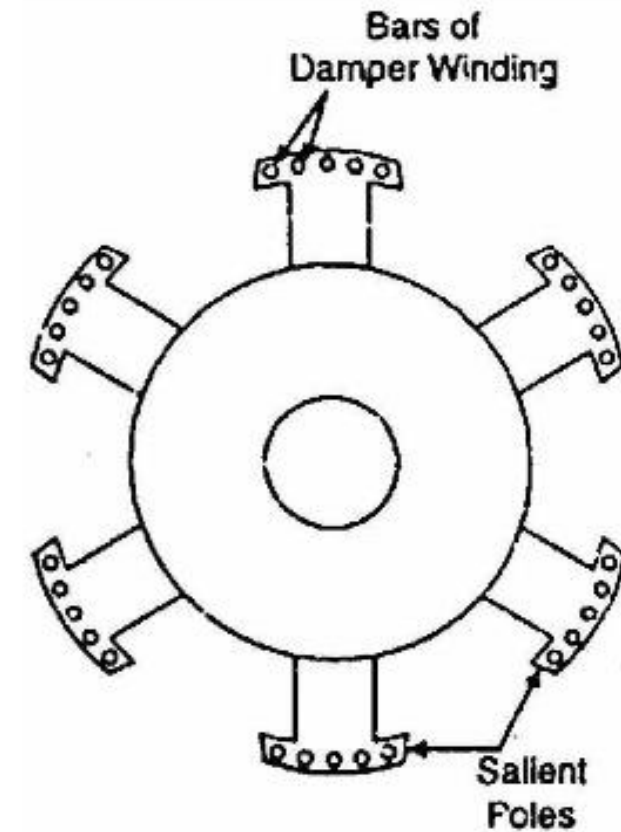


Fig. 12 Winding diagram

Assignment: Draw the developed winding diagram of a 3 phase induction motor with 18 slots, 2 poles, single layer, short pitched winding by 1 slot

Hunting



- (i) Hunting generally occurs in alternators driven by engines because the driving torque of engines is not uniform.
- (ii) Alternators driven by steam turbines generally do not have a tendency to hunt since the torque applied does not pulsate.
- (iii) In cylindrical rotor machines, the damper windings are generally not used. It is because the solid rotor provides considerable damping.

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Ref [1]: The evolution of the synchronous machine by Professor Gerhard Neidhofer

DOI: [10.1049/esej:19920050](https://doi.org/10.1049/esej:19920050). URL: <https://ieeexplore.ieee.org/document/177464>

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2. <https://new.siemens.com/global/en/products/energy/power-generation/generators/s-gen-100a.html>
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Youtube Links:

1. <https://www.youtube.com/watch?v=tiKH48EMgKE>
2. <https://www.youtube.com/watch?v=Ak6Zf-wJpsM>
3. <https://www.youtube.com/user/ABBmotorsgenerators>
4. <https://www.youtube.com/watch?v=kOwMZ6axdSk>
5. <https://www.youtube.com/watch?v=pfj4w2Jz3xA>